

Chapter 5 - Two Dimensional Analytical Geometry

Top of Form

Top of Form

Ex5.1

Question 1.

Obtain the equation of the circles with radius 5 cm and touching x-axis at the origin in general form.

Solution:

Given radius = 5 cm and the circle is touching x axis

So centre will be $(0, \pm 5)$ and radius = 5

The equation of the circle with centre $(0, \pm 5)$ and radius 5 units is $(x-0)^2 + (y \pm 5)^2 = 5^2$

(i.e) $x^2 + y^2 \pm 10y + 25 - 25 = 0$

(i.e) $x^2 + y^2 \pm 10y = 0$

Question 2.

Find the equation of the circle with centre $(2, -1)$ and passing through the point $(3, 6)$ in standard form.

Solution:

Centre = $C = (2, -1)$; Passing through = $A = (3, 6)$

So radius = $CA = \sqrt{(2-3)^2 + (-1-6)^2} = \sqrt{1+49} = 5\sqrt{50}$

Now centre = $(2, -1)$ and radius = $\sqrt{50}$

So equation of the circle is

(i.e) $(x-2)^2 + (y+1)^2 = 50 \Rightarrow (x-2)^2 + (y+1)^2 = 50$

Question 3.

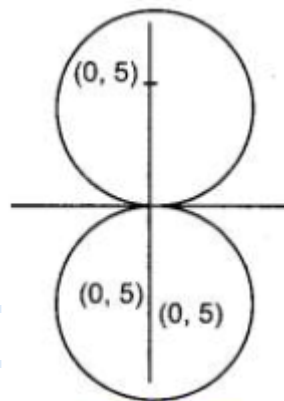
Find the equation of circles that touch both the axes and pass through $(-4, -2)$ in general form.

Solution:

Since the circle touches both the axes, its centre will be (r, r) and radius will be r .

Here centre $= C = (r, r)$ and point on the circle is $A = (-4, -2)$

$$CA = r \Rightarrow CA^2 = r^2$$



(i.e) $(r+4)^2 + (r+2)^2 = r^2 \Rightarrow r^2 + 8r + 16 + r^2 + 4r + 4 - r^2 = 0$ (i.e) $r^2 + 12r + 20 = 0 \Rightarrow (r+2)(r+10) = 0 \Rightarrow r = -2$ or -10

When $r = -2$, the equation of the circle will be $(x+2)^2 + (y+2)^2 = 22$

(i.e) $x^2 + y^2 + 4x + 4y + 4 = 0$

When $r = -10$, the equation of the circle will be $(x+10)^2 + (y+10)^2 = 102$

(i.e) $x^2 + y^2 + 20x + 20y + 100 = 0$

Question 4.

Find the equation of the circle with centre $(2, 3)$ and passing through the intersection of the lines $3x - 2y - 1 = 0$ and $4x + y - 27 = 0$

Solution:

To find the point of intersection of the two lines we have to solve the two equations.

Now solving them: $3x - 2y = 1$ $4x + y = 27$(2)

(2) $\times 2 \Rightarrow 8x + 2y = 54$

(1) $\Rightarrow 3x - 2y = 1$

(3) $+(1) \Rightarrow 11x = 55 \Rightarrow x = 5$

Substituting $x = 5$ in (2) we get

$20 + y = 27 \Rightarrow y = 27 - 20 = 7$

$\therefore$ The point $= A = (5, 7)$

Given centre $= C = (2, 3)$

$\therefore$ radius $= \sqrt{(5-2)^2 + (7-3)^2} = \sqrt{9+16} = 5$

Now centre $= (2, 3)$ and radius $= 5$

So equation of the circle is

$$(x-2)^2 + (y-3)^2 = 52$$

$$(i.e) x^2 + y^2 - 4x - 6y - 12 = 0$$

Question 5.

Obtain the equation of the circle for which (3,4) and (2,-7) are the ends of a diameter.

Solution:

The equation of a circle with (x_1, y_1) and (x_2, y_2) as end points of a diameter is $(x-x_1)(x-x_2) + (y-y_1)(y-y_2) = 0$

Here the end points of a diameter are (3,4) and (2,-7)

So equation of the circle is $(x-3)(x-2) + (y-4)(y+7) = 0$

$$x^2 + y^2 - 5x + 37 - 22 = 0$$

Question 6.

Find the equation of the circle through the points (1,0), (-1,0), and (0,1).

Solution:

Let the required circle be

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

The circle passes through (1,0), (-1,0) and (0,1)

$$(1,0) \Rightarrow 1 + 0 + 2g(1) + 2f(0) + c = 0 \Rightarrow 2g + c = -1 \dots (1)$$

$$(-1,0) \Rightarrow 1 + 0 + 2g(-1) + 2f(0) + c = 0$$

$$-2g + c = -1 \dots (2)$$

$$(0,1) \Rightarrow 0 + 1 + 2g(0) + 2f(1) + c = 0 \Rightarrow 2f + c = -1 \dots (3)$$

$$\text{Now solving (1), (2) and (3) } 2g + c = -1 \dots (1) \quad -2g + c = -1 \dots (2) \quad (1) + (2) \Rightarrow 2c = -2 \Rightarrow c = -1$$

Substituting $c = -1$ in (1) we get

$$2g - 1 = -1 \Rightarrow 2g = 0 \Rightarrow g = 0$$

Substituting $c = -1$ in (3) we get

$$2f - 1 = -1 \Rightarrow 2f = 0 \Rightarrow f = 0$$

So we get $g = 0, f = 0$ and $c = -1$

So the required circle will be

$$x^2 + y^2 + 2(0)x + 2(0)y - 1 = 0 \quad (i.e) \quad x^2 + y^2 - 1 = 0 \Rightarrow x^2 + y^2 = 1$$

Question 7.

A circle of area 9π square units has two of its diameters along the lines $x+y=5$ and $x-y=1$. Find the equation of the circle.

Solution:

Area of the circle $= 9\pi$

$$(i.e) \pi r^2 = 9\pi \Rightarrow r^2 = 9 \Rightarrow r = 3$$

(i.e) radius of the circle $= r = 3$

The two diameters are $x+y=5$ and $x-y=1$

The point of intersection of the diameter is the centre of the circle $= C$

To find C : Solving $x+y=5$ $x-y=1$

$$(1) + (2) \Rightarrow 2x = 6 \Rightarrow x = 3$$

Substituting $x=3$ in (1) we get

$$3+y=5 \Rightarrow y=5-3=2$$

$\therefore$ Centre $= (3,2)$ and radius $= 3$

So equation of the circle is $(x-3)^2 + (y-2)^2 = 3^2$

$$(i.e) x^2 + y^2 - 6x - 4y + 4 = 0$$

Question 8.

If $y = 2\sqrt{2}x + c$ is a tangent to the circle $x^2 + y^2 = 16$, find the value of c .

Solution:

The condition for the line $y = mx + c$ to be a tangent to the circle $x^2 + y^2 = a^2$ is

$$c^2 = a^2(1+m^2)$$

$$\text{Here } x^2 + y^2 = 16 \Rightarrow a^2 = 16$$

$$y = 2\sqrt{2}x + c \Rightarrow m = 2\sqrt{2} \text{ and } c = c$$

The condition is $c^2 = a^2(1+m^2)$

$$(i.e) c^2 = 16(1+8) = 144$$

$$\Rightarrow c = \pm 12$$

Question 9.

Find the equation of the tangent and normal to the circle $x^2 + y^2 - 6x + 6y - 8 = 0$ at $(2,2)$.

Solution:

The equation of the tangent to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ at (x_1, y_1) is

$$xx_1 + yy_1 + g(x+x_1) + f(y+y_1) + c = 0$$

So the equation of the tangent to the circle $x^2 + y^2 - 6x + 6y - 8 = 0$ at (x_1, y_1) is

$$xx_1 + yy_1 - 3(x+x_1) + 3(y+y_1) - 8 = 0 \text{ (i.e) } xx_1 + yy_1 - 3(x+x_1) + 3(y+y_1) - 8 = 0 \text{ Here } (x_1, y_1) = (2, 2)$$

So equation of the tangent is

$$x(2) + y(2) - 3(x+2) + 3(y+2) - 8 = 0$$

$$(i.e) 2x + 2y - 3x - 6 + 3y + 6 - 8 = 0$$

$$(i.e) -x + 5y - 8 = 0 \text{ or } x - 5y + 8 = 0$$

Normal is a line $\perp$ to the tangent

So equation of normal circle be of the form $5x+y+k=0$

The normal is drawn at (2,2)

$$\Rightarrow 10+2+k=0 \Rightarrow k=-12$$

So equation of normal is $5x+y-12=0$

Question 10.

Determine whether the points $(-2,1)$, $(0,0)$ and $(-4,-3)$ lie outside, on or inside the circle $x^2+y^2-5x+2y-5=0$.

Solution:

To find the position of a point with regard to a given circle, substitute the point in the equation of the circle if we get a positive value, the point lies outside the circle.

If we get a -ve value the point lies inside the circle and if we get 0 then the point lies on the circumference of the circle.

The given circle is $x^2+y^2-5x+2y-5=0$

Substituting the point $(-2,1)$ in (1) we get

$$4+1-5(-2)+2(1)-5=5+10+2-5=12$$

$\Rightarrow (-2,1)$ lies outside the circle

Substituting the point $(0,0)$ in (1) we get

$-5 < 0 \Rightarrow (0,0)$ lies inside the circle Substituting the point $(-4,-3)$ in (1) we get

$$16+9+20-6-5=34 > 0$$

$\Rightarrow (-4,-3)$ lies outside the circle

Physics

Question 11.

Find centre and radius of the following circles.

(i) $x^2+(y+2)^2=0$

(ii) $x^2+y^2+6x-4y+4=0$

(iii) $x^2+y^2-x+2y-3=0$

(iv) $2x^2+2y^2-6x+4y+2=0$

Solution:

(i) $x^2+(y+2)^2=0$

(i.e) $x^2+y^2+4y+4=0$

Comparing this equation with the general form $x^2+y^2+2gx+2fy+c=0$

we get $2g=0 \Rightarrow g=0$

$$2f=4 \Rightarrow f=2 \text{ and } c=4$$

$$\text{Now centre } =(-g,-f)=(0,-2)$$

$$\text{Radius } =r=\sqrt{g^2+f^2-c}=\sqrt{0+4-4} \therefore \text{Centre } =(0,-2) \text{ and radius } =0$$

$$(ii) x^2+y^2+6x-4y+4=0$$

Comparing with the general form we get

$$2g=6, 2f=-4 \Rightarrow g=3, f=-2 \text{ and } c=4 \text{ Centre } =(-g,-f)=(-3,2) \text{ Radius } =\sqrt{g^2+f^2-c}=\sqrt{9+4-4}=3 \therefore$$

$$\text{Centre } =(-3,2) \text{ and radius } =3$$

$$(iii) x^2+y^2-x+2y-3=0$$

Comparing with the general form of the circle we get

$$2g=-1, 2f=2, c=-3$$

$$\text{So centre } =(-g,-f)=(1/2, -1) \text{ Radius } =\sqrt{g^2+f^2-c}=\sqrt{1/4+1+3}=\sqrt{17/4} \therefore \text{Centre } =(1/2, -1) \text{ and radius } =\sqrt{17}/2$$

$$(iv) 2x^2+2y^2-6x+4y+2=0$$

$$(\div \text{ by } 2) \Rightarrow x^2+y^2-3x+2y+1=0$$

Comparing this equation with the general form of the circle we get

$$2g=-3, 2f=2$$

$$g=-3/2, f=1 \text{ and } c=1$$

$$\text{So centre } =(-g,-f)=(3/2, -1)$$

$$\text{and radius } =\sqrt{g^2+f^2-c}=\sqrt{9/4+1-1}=3/2$$

$$\therefore \text{Centre } =(3/2, -1) \text{ and radius } =3/2$$

Question 12.

Education

If the equation $3x^2 + (3-p)xy + qy^2 - 2px = 8pq$ represents a circle, find p and q . Also determine the centre and radius of the circle.

Solution:

For a circle co-eff of x^2 = co-eff of y^2

$$\Rightarrow 3=q$$

co-eff of $xy=0$

$$\Rightarrow 3-p=0 \Rightarrow p=3$$

$$\text{So } p=q=3$$

So the equation of the circle becomes $3x^2+3y^2-6x-72=0$

$$(\div \text{ by } 3) \Rightarrow x^2+y^2-2x-24=0$$

Comparing this equation with the general form of the circle we get

$$2g = -2, 2f = 0 \Rightarrow g = -1, f = 0 \text{ and } c = -24$$

So centre $= (-g, -f) = (1, 0)$ and radius $= \sqrt{g^2 + f^2 - c} = 5$

$\therefore$ Centre $= (1, 0)$ and radius $= 5$

CONTACT US

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Ex 5.2

Question 1.

Find the equation of the parabola in each of the cases given below:

- (i) focus $(4, 0)$ and directrix $x = -4$.
- (ii) passes through $(2, -3)$ and symmetric about y-axis.
- (iii) vertex $(1, -2)$ and focus $(4, -2)$.
- (iv) end points of latus rectum $(4, -8)$ and $(4, 8)$.

Solution:

(i) Focus $= F(4, 0)$

$$\Rightarrow a = 4$$

Equation of directrix $x = -4$

$\Rightarrow$ The curve open to the right. So the equation will be of the form $y^2 = 4ax$ Here $a = 4$

$$\Rightarrow y^2 = 4(4)x \text{ (i.e.) } y^2 = 16x$$

(ii) The parabola is symmetric about y axis. So the equation will be of the form $x^2 = 4ay$

It passes through $(2, -3)$

$$\Rightarrow 2^2 = 4a(-3) \Rightarrow -12a \Rightarrow a = -\frac{1}{3} \Rightarrow 4a = -\frac{4}{3} \therefore \text{Equation of parabola is } x^2 = -\frac{4}{3}y$$

(iii) The distance between vertex and focus $= 3$

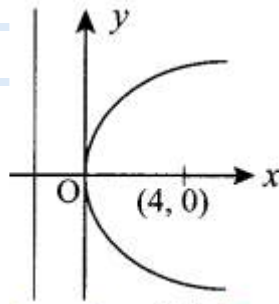
(i.e.,) $a = 3$

Parabola is open to the right.

So equation will be of the form $y^2 = 4ax$

$$\text{Here } a = 3 \Rightarrow y^2 = 12x$$

but the vertex is $(1, -2)$



So equation of the parabola is

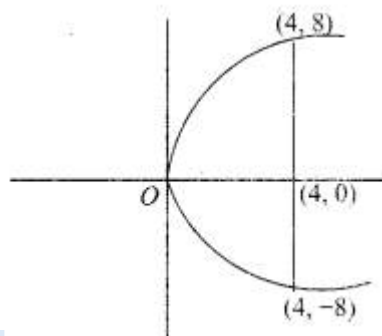
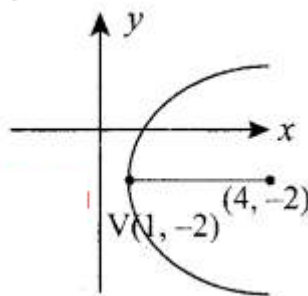
$$(y+2)^2=12(x-1)$$

(iv) Focus =(4,0)

Equation of the parabola will be of the form . $y^2=4ax$

Here $a=4$

$$\Rightarrow y^2=16x$$



Question 2.

Find the equation of the ellipse in each of the cases given below:

(i) foci $(\pm 3, 0)$, $e=1/2$

(ii) foci $(0, \pm 4)$ and end points of major axis are $(0, \pm 5)$.

(iii) length of latus rectum 8 , eccentricity $=3/5$ and major axis on x-axis.

(iv) length of latus rectum 4 , distance between foci $4\sqrt{2}$ and major axis as y-axis.

Solution:

(i) Given $ae = 3$ and $e = 12$

$$\Rightarrow a(12) = 3 \Rightarrow a = 6$$

$$\text{So } a^2 = 36$$

$$b^2 = a^2(1 - e^2) = 36(1 - 144) = 36 \times 34 = 27$$

Since Foci $= (\pm 3, 0)$, major axis is along x-axis

So equation of ellipse is $\frac{x^2}{36} + \frac{y^2}{27} = 1$

(ii) From the diagram we see that major axis is along y-axis.

Also $a = 5$ and $ae = 4$

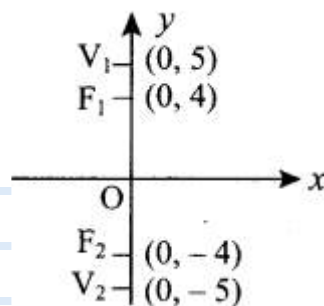
$$\Rightarrow 5e = 4 \Rightarrow e = \frac{4}{5}$$

$$\text{Now } a = 5 \Rightarrow a^2 = 25$$

$$ae = 4 \Rightarrow a^2e^2 = 16$$

$$\text{We know } b^2 = a^2(1 - e^2) = a^2 - a^2e^2 = 25 - 16 = 9$$

Equation of ellipse is $\frac{x^2}{9} + \frac{y^2}{25} = 1$



(iii) Given $2b^2a = 8$ and $e = \frac{3}{5}$

Now

$$\text{But } b^2 = a^2(1 - e^2) \Rightarrow 4a = a^2(1 - (\frac{3}{5})^2) = a^2(1 - \frac{9}{25})$$

$$2b^2a = 8 \Rightarrow b^2 = \frac{8}{2a} = \frac{4}{a}$$

$$\Rightarrow 4a = 16a^2 \Rightarrow a = \frac{100}{16} = \frac{25}{4}$$

So

$$a^2 = \frac{625}{16} \text{ and } b^2 = \frac{4}{a} = \frac{4 \times 25}{16} = \frac{25}{4}$$

Here major axis is along x-axis

Equation of ellipse is

$$\frac{x^2}{\frac{625}{16}} + \frac{y^2}{\frac{25}{4}} = 1 \Rightarrow \frac{16x^2}{625} + \frac{y^2}{\frac{25}{4}} = 1$$

(iv) Given $2b^2a = 4$ and $2ae = 4\sqrt{2}$

$$\text{Now } 2b^2a = 4 \Rightarrow b^2 = \frac{2}{a}$$

$$\Rightarrow b^2 = \frac{2}{a} \Rightarrow \frac{2}{a} = \frac{4\sqrt{2}}{2a} \Rightarrow a = 2\sqrt{2}$$

$$\text{So } a^2e^2 = 4(2) = 8$$

$$\text{We know } b^2 = a^2(1 - e^2) = a^2 - a^2e^2$$

$$\Rightarrow 2a = a^2 - 8 \Rightarrow a^2 - 2a - 8 = 0 \Rightarrow (a-4)(a+2) = 0 \Rightarrow a = 4 \text{ or } -2$$

As a cannot be negative

$$a = 4 \text{ So } a^2 = 16 \text{ and } b^2 = 2(4) = 8$$

Also major axis is along y -axis

$$\text{So equation of ellipse is } x^2/8 + y^2/16 = 1$$

Question 3.

Find the equation of the hyperbola in each of the cases given below:

(i) foci $(\pm 2, 0)$, eccentricity $= 3/2$

(ii) Centre $(2, 1)$, one of the foci $(8, 1)$ and corresponding directrix $x = 4$.

(iii) passing through $(5, -2)$ and length of the transverse axis along YC axis and of length 8 units.

Solution:

(i) Given

$$ae = 2 \text{ and } e = 3/2 \Rightarrow a(3/2) = 2 \Rightarrow a = 4/3 \text{ So } a^2 = 16/9 \text{ } b^2 = a^2(e^2 - 1) = a^2e^2 - a^2 = 4 - 16/9 = 20/9$$

Since the foci are $(\pm 2, 0)$, transverse axis is along x -axis

So equation of hyperbola is

$$x^2/16/9 - y^2/20/9 = 1 \Rightarrow 9x^2/16 - 9y^2/20 = 1$$

(ii) Given Centre $= (2, 1)$

$$ae = 6 \text{ (distance between } (2, 1) \text{ and } (8, 1))$$

$$\text{Also } ae = 2 \Rightarrow a = 2e$$

$$\text{Equation of directrix is } x = 4 \text{ [(i.e.,) } (x - 2) = 2 \text{ Since centre is } (2, 1) \text{] } \Rightarrow ae = 2$$

$$\text{Given } ae = 6 \Rightarrow a^2e^2 = 36$$

$$\text{(i.e.) } (2e)^2(e)^2 = 36 \Rightarrow 4e^4 = 36 \Rightarrow e^4 = 9 \Rightarrow e = \sqrt{3}$$

$$\text{Now } e = \sqrt{3} \Rightarrow a = 2\sqrt{3}$$

$$\therefore a^2 = 4 \times 3 = 12 \text{ } b^2 = a^2(e^2 - 1) = a^2e^2 - a^2 = 36 - 12 = 24$$

So here Centre $= (2, 1)$

So equation of hyperbola is

$$(x - 2)^2/12 - (y - 1)^2/24 = 1$$

(iii) Length of the transverse axis $= 8$

$$2a = 8 \Rightarrow a = 4$$

Transverse axis is along x -axis

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So of equation of hyperbola is will be

$$x^2/a^2 - y^2/b^2 = 1 \text{ (i.e.,) } x^2/16 - y^2/b^2 = 1$$

But the hyperbola passes through (5,-2)

$$\Rightarrow 2516 - 4b^2 = 1 \Rightarrow 2516 = 1 + 4b^2 \text{ (or) } 2516 - 1 = 4b^2 \Rightarrow 4b^2 = 916 \Rightarrow b^2 = 649$$

$$x^2 - y^2 - 649 = 1 \Rightarrow x^2 - y^2 - 9y^2 - 64 = 1$$

Question 4.

Find the vertex, focus, equation of directrix and length of the latus rectum of the following:

(i) $y^2 = 16x$

(ii) $x^2 = 24y$

(iii) $y^2 = -8x$

(iv) $x^2 - 2x + 8y + 17 = 0$

(v) $y^2 - 4y - 8x + 12 = 0$

Solution:

(i) $y^2 = 16x$

It is of the form $y^2 = 4ax$ (type I)

Here $4a = 16 \Rightarrow a = 4$

Vertex = (0,0)

Focus = (a,0) = (4,0)

Equation of directrix $x + 4 = 0$ (or) $x = -4$

Length of latus rectum = $4a = 16$.

(ii) $x^2 = 24y$

This is of the form $x^2 = 4ay$ (type III) $4a = 24 \Rightarrow a = 6$

Vertex = (0,0)

Focus = (0,a) = (0,6)

Equation of directrix is $y + a = 0$ (i.e.,) $y + 6 = 0$ (or) $y = -6$

Length of latus rectum = $4a = 24$.

(iii) $y^2 = -8x$

This is of the form $y^2 = -4ax$ (type II)

Here $4a = 8 \Rightarrow a = 2$

Vertex = (0,0)

Focus = (-a,0) = (-2,0)

Equation of directrix is $x - 2 = 0$ (or) $x = 2$

Length of latus rectum = $4a = 8$.

(iv)

$$x^2 - 2x + 8y + 17 = 0 \Rightarrow x^2 - 2x = -8y - 17 \Rightarrow x^2 - 2x + 1 - 1 = -8y - 17 \Rightarrow (x-1)^2 - 1 = -8y - 17 \Rightarrow (x-1)^2 = -8(y-2)$$

Taking $x-1=X$ and $y-2=Y$.

We get $X^2 = -8Y$.

This is of the form $x^2 = -4ay$ (type IV)

Where $4a=8 \Rightarrow a=2$

	Referred to X, Y axes	Referred to x, y axes
Vertex	$(0, 0)$	$(1, -2)$
Focus	$(0, -a) = (0, -2)$	$(0, -2) + (1, -2) = (1, -4)$
Equation of directrix	$Y - 2 = 0$	$y + 2 - 2 = 0$ (ie.,) $y = 0$
Length of latus rectum	$4a = 8$	$4a = 8$

$$(v) y^2 - 4y = 8x - 12 \Rightarrow y^2 - 4y + 4 = 8x - 12 + 4 \Rightarrow (y-2)^2 = 8x - 8 = 8(x-1)$$

Taking $x-1=X$ and $y-2=Y$.

We get $Y^2 = 8X$.

This is of the form $y^2 = 4ax$ (type IV)

Where $4a=8 \Rightarrow a=2$

	Referred to X, Y axes	Referred to x, y axes
Vertex	$(0, 0)$	$(1, 2)$
Focus	$(a, 0) = (2, 0)$	$(2, 0) + (1, 2) = (3, 2)$
Equation of directrix	$X + 2 = 0$	$x - 1 + 2 = 0 \Rightarrow x + 1 = 0$
Length of latus rectum	$4a = 8$	$4a = 8$

Question 5.

Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:

(i) $x^2 + 25y^2 = 1$

(ii) $x^2 + 23y^2 = 1$

(iii) $x^2 - 25y^2 = 1$

(iv) $y^2 - x^2 = 1$

Solution:

(i) $x^2 + 25y^2 = 1$

It is of the form $x^2/a^2 + y^2/b^2 = 1$, which is an ellipse

Here $a^2 = 25$, $b^2 = 9$

$a = 5$, $b = 3$

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{9}{25} = \frac{16}{25} \Rightarrow e = \frac{4}{5}$$

Now $e = \frac{4}{5}$ and $a = 5 \Rightarrow ae = 4$ and $ae = 5 \cdot \frac{4}{5} = 4$

Here the major axis is along x axis

$\therefore$ Centre $= (0,0)$

Foci $= (\pm ae, 0) = (\pm 4, 0)$

Vertices $= (\pm a, 0) = (\pm 5, 0)$

Equation of directrix $x = \pm ae$ (ie.,) $x = \pm 25/4$

(ii) $x^2/25 + y^2/10 = 1$

It is an ellipse and here (always $a > b$)

Here

$$a^2 = 25, b^2 = 10 \Rightarrow e^2 = \frac{a^2 - b^2}{a^2} = \frac{25 - 10}{25} = \frac{15}{25} \Rightarrow e = \frac{\sqrt{15}}{5}$$

Now

Here the major axis is along y-axis

Centre $= (0,0)$

Foci $= (0, \pm ae) = (0, \pm \sqrt{15})$

Vertices $= (0, \pm a) = (0, \pm 5)$

Equation of directrices $y = \pm 10/\sqrt{15}$

(iii) $x^2/25 - y^2/144 = 1$

This is a hyperbola of the form $x^2/a^2 - y^2/b^2 = 1$.

Here

$$a^2 = 25, b^2 = 144 \Rightarrow e^2 = \frac{a^2 + b^2}{a^2} = \frac{25 + 144}{25} = \frac{169}{25} \Rightarrow e = \frac{13}{5}$$

Now $e = 13/5$ and $a = 5 \Rightarrow ae = 13$ and $ae = 5 \cdot 13/5 = 13$

Now, Here transverse axis is along x-axis

Centre $= (0,0)$

Vertices $= (\pm a, 0) = (\pm 5, 0)$

Foci $= (\pm ae, 0) = (\pm 13, 0)$

Equation of directrices $x = \pm ae$ (ie.,) $x = \pm 25/13$ (iv) $y^2/16 - x^2/9 = 1$

It is a hyperbola. Here transverse axis is along y-axis ..

Also $a^2 = 16$ and $b^2 = 9$

Now

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{16 + 9}{16} = \frac{25}{16} \Rightarrow e = \frac{5}{4}$$

$e = 5/4$ and $a = 4 \Rightarrow ae = 5$ and $ae = 4 \cdot 5/4 = 5$

Now Centre $= (0,0)$

Vertices $= (0, \pm a) = (0, \pm 4)$

Foci $= (0, \pm a) = (0, \pm 5)$

Equation of directrices $y = \pm 16/5$

Question 6.

Prove that the length of the latus rectum of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ is $2b^2/a$

Solution:

The latus rectum LL' of the hyperbola $x^2/a^2 - y^2/b^2 = 1$ passes through the focus $(ae, 0)$

So, Let $L = (ae, y)$ (say)

$$a^2/a^2 - y^2/b^2 = 1 \Rightarrow y^2/b^2 = e^2 - 1 \Rightarrow y^2 = b^2(e^2 - 1) = b^2(b^2/a^2) (\because e^2 = 1 + b^2/a^2) \Rightarrow y = \pm b^2/a$$

That is the end point of latus rectum L and L' are $(ae, b^2/a)$ and $(ae, -b^2/a)$.

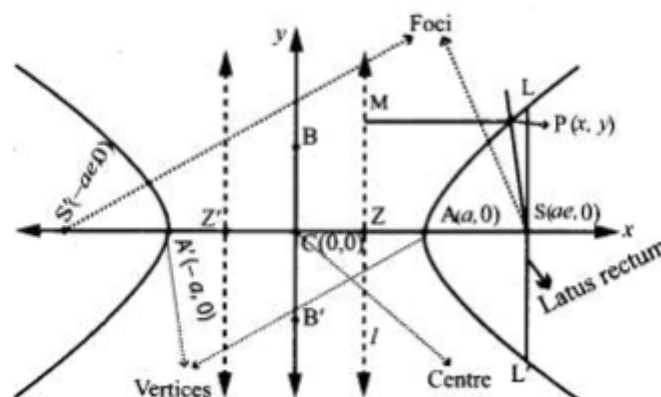
So the length of the latus rectum $L'L = 2b^2/a$.

Question 7.

Show that the absolute value of the difference of the focal distances of any point P on the hyperbola is the length of its transverse axis.

Solution:

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We know $SP/PM = e \Rightarrow SP = ePM \Rightarrow SP = e(x - ae) = ex - a$ Again

$$S'P/PM' = e \Rightarrow S'P = ePM' = e(ae + x) = a + ex$$

$$\therefore S'P - SP = (a + ex) - (ex - a) = a + ex - ex + a = 2a \text{ (transverse axis)}$$

Question 8.

Identify the type of conic and find centre, foci, vertices, and directrices of each of the following:

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(i) $(x-3)^2/25 + (y-4)^2/89 = 1$

(ii) $(x+1)^2/100 + (y-2)^2/64 = 1$

(iii) $(x+3)^2/25 - (y-4)^2/64 = 1$

$$(iv) (y-2)^2 - 25 - (x+1)^2 - 16 = 1$$

$$(v) 18x^2 + 12y^2 - 144x + 48y + 120 = 0$$

$$(vi) 9x^2 - y^2 - 36x - 6y + 18 = 0$$

Solution:

$$(i) (x-3)^2 - 25 + (y-4)^2 - 289 = 1$$

Now taking $x-3=X$ and $y-4=Y$ We get

$$X^2 - 25 + Y^2 - 289 = 1$$

It is of the form $x^2/a^2 + y^2/b^2 = 1$, Which is an ellipse

Here major axis is along Y-axis also $a > b$

$$\Rightarrow a^2 = 289 \text{ and } b^2 = 25 (a > b) \Rightarrow a = 17$$

Now

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{25}{289} = \frac{264}{289}$$

$$\Rightarrow e = \frac{16}{17}$$

$$\text{Now } e = \frac{16}{17}, a = 17$$

$$\Rightarrow ae = 16 \text{ and } ae = 17 \cdot \frac{16}{17} = 16$$

	Referred to X, Y axes	Referred to x, y axes
Centre	(0, 0)	(3, 4)
Vertices	$(0, \pm a) = (0, \pm 17)$	$V_1 = (0, 17) + (3, 4) = (3, 21)$ $V_2 = (0, -17) + (3, 4) = (3, -13)$
Foci	$(0, \pm ae) = (0, \pm 16)$	$F_1 = (0, 16) + (3, 4) = (3, 20)$ $F_2 = (0, -16) + (3, 4) = (3, -12)$
Equation of directrix	$Y = \pm \frac{a}{e} = \pm \frac{289}{16}$	$y - 4 = \pm \frac{289}{16}$ $y = + \frac{289}{16} + 4 = \frac{321}{16}$ or $y = - \frac{289}{16} + 4 = - \frac{257}{16}$

$$(ii) (x+1)^2 - 100 + (y-2)^2 - 64 = 1$$

Taking $x+1=X$ and $y-2=Y$ We get

$$X^2 - 100 + Y^2 - 64 = 1$$

It is of the form $x^2/a^2 + y^2/b^2 = 1$, Which is an ellipse

Here $a^2 = 100$ and $b^2 = 64 (a > b)$

$$\Rightarrow a = 10 \text{ and } b = 8$$

Now

$$e^2 = 1 - \frac{b^2}{a^2} = 1 - \frac{64}{100} = \frac{36}{100} = \frac{9}{25}$$

$$\Rightarrow e=35$$

Now $a=10$ and $e=35, ae=10 \times 35=6$ and $ae=103/5=503$

Here major axis is along X-axis

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	Referred to X, Y axes	Referred to x, y axes
Centre	$(0, 0)$	$(-1, 2)$
Foci	$(\pm a, 0) = (\pm 6, 0)$	$F_1 = (6, 0) + (-1, 2) = (5, 2)$ $F_2 = (-6, 0) + (-1, 2) = (-7, 2)$
Vertices	$(\pm a, 0) = (\pm 10, 0)$	$V_1 = (10, 0) + (-1, 2) = (9, 2)$ $V_2 = (-10, 0) + (-1, 2) = (-11, 2)$
Equation of directrix	$X = \pm \frac{a}{e} = \pm \frac{50}{3}$	$x + 1 = \pm \frac{50}{3}$ $x = + \frac{50}{3} - 1$ (ie.,) $x = \frac{47}{3}$ (or) $x = \frac{-50}{3} - 1$ (ie.,) $x = \frac{-53}{3}$

$$(iii) (x+3)^2 - (y-4)^2 = 1$$

Taking $x+3=X$ and $y-4=Y$ We get $X^2 - Y^2 = 1$

It is of the form $x^2/a^2 - y^2/b^2 = 1$, Which is a hyperbola

(ie.,) The given conic is a hyperbola

Here $a^2=225, b^2=64$

$\Rightarrow a=15$ Now $e^2 = a^2 + b^2 = 225 + 64 = 289 \Rightarrow e = \sqrt{289} = 17$ Now $a=15, e=17, ae=17$ and $ae=1517/15=22517$

Here the transverse axis is along X-axis

	Referred to X, Y axes	Referred to x, y axes
Centre	$(0, 0)$	$(-3, 4)$
Foci	$(\pm ae, 0) = (\pm 17, 0)$	$F_1 = (17, 0) + (-3, 4) = (14, 4)$ $F_2 = (-17, 0) + (-3, 4) = (-20, 4)$
Vertices	$(\pm a, 0) = (\pm 15, 0)$	$V_1 = (15, 0) + (-3, 4) = (12, 4)$ $V_2 = (-15, 0) + (-3, 4) = (-18, 4)$
Equation of directrix	$X = \pm \frac{a}{e} = \pm \frac{225}{17}$	$x + 3 = \pm \frac{225}{17}$ $x = + \frac{225}{17} - 3 = \frac{174}{17}$ $x = \frac{-225}{17} - 3 = \frac{-276}{17}$

(iv) $(y-2)^2 - (x+1)^2 = 1$

Taking $y-2=Y$ and $x+1=X$ We get

$$Y^2 - X^2 = 1$$

It is of the form $y^2/a^2 - x^2/b^2 = 1$, Which is a hyperbola

Here the transverse axis is along Y-axis

Also $a^2=25, b^2=16$

$$\Rightarrow a=5$$

Now $e^2 = a^2 + b^2/a^2 = 25 + 16/25 = 41/25$

$$\Rightarrow e = \sqrt{41}/5$$

Now

$$a=5, e=\sqrt{41}/5, ae=\sqrt{41} \text{ and } ae=5\sqrt{41}/5=\sqrt{41}$$

(v) $18x^2 + 12y^2 - 144x + 48y + 120 = 0$
 $(18x^2 - 144x) + (12y^2 + 48y) = -120$
 $18(x^2 - 8x) + 12(y^2 + 4y) = -120$
 $18(x^2 - 8x + 16 - 16) + 12(y^2 + 4y + 4 - 4) = -120$
 $18(x-4)^2 - 288 + 12(y+2)^2 - 48 = -120$
 $18(x-4)^2 + 12(y+2)^2 = 120$
 $(x-4)^2/20 + (y+2)^2/10 = 1$

(+ by 216) $\Rightarrow 18(x-4)^2 + 12(y+2)^2 = 120 \Rightarrow (x-4)^2/20 + (y+2)^2/10 = 1$

Taking $x-4=X$ and $y+2=Y$

We get $X^2/20 + Y^2/10 = 1$

It is of the form $x^2/a^2 + y^2/b^2 = 1$, Which is an ellipse

Here the major axis is along Y-axis

Also $a^2=18, b^2=12 (a>b)$
 $a=3\sqrt{2}, b=2\sqrt{3}$
 $e^2 = a^2 - b^2/a^2 = 18 - 12/18 = 16/18 = 8/9$
 $e=2\sqrt{2}/3$
 $ae=3\sqrt{2} \cdot 2\sqrt{2}/3 = 4$
 $ae=3\sqrt{2}/\sqrt{3} = \sqrt{6}$

	Referred to X, Y axes	Referred to x, y axes
Centre	(0, 0)	(4, -2)
Foci	$(0, \pm ae) = (0, \pm \sqrt{6})$	$F_1 = (4, -2) + (0, \sqrt{6}) = (4, -2 + \sqrt{6})$ $F_2 = (4, -2) + (0, -\sqrt{6}) = (4, -2 - \sqrt{6})$
Vertices	$(0, \pm a) = (0, \pm 3\sqrt{2})$	$V_1 = (4, -2) + (0, 3\sqrt{2}) = (4, -2 + 3\sqrt{2})$ $V_2 = (4, -2) + (0, -3\sqrt{2}) = (4, -2 - 3\sqrt{2})$
Equation of directrices	$Y = \pm \frac{a}{e} = \pm 3\sqrt{6}$	$y + 2 = \pm 3\sqrt{6}$ $y = -2 + 3\sqrt{6}$ $y = -2 - 3\sqrt{6}$

(vi)

$9x^2 - y^2 - 36x - 6y + 18 = 0$
 $(9x^2 - 36x) - (y^2 + 6y) = -18$
 $9(x^2 - 4x) - (y^2 + 6y) = -18$
 $9(x^2 - 4x + 4 - 4) - (y^2 + 6y + 9 - 9) = -18$
 $9(x-2)^2 - 36 - (y+3)^2 + 9 = -18$
 $9(x-2)^2 - (y+3)^2 = 18$
 $(x-2)^2/2 - (y+3)^2/2 = 1$

Taking $x-2=X$ and $y+3=Y$

We get

$$X^2 - Y^2 = 9$$

It is of the form $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ Which is a hyperbola,

The transverse axis is along X-axis. Here $a^2=1, b^2=9 \Rightarrow a=1$

$$e^2 = \frac{a^2 + b^2}{a^2} = \frac{1+9}{1} = 10 \Rightarrow e = \sqrt{10} \text{ Now } e = \sqrt{10} \text{ and } a=1, ae = \sqrt{10} \text{ and } ae = 1/\sqrt{10}$$

	Referred to X, Y axes	Referred to x, y axes
Centre	(0, 0)	(2, -3)
Foci	$(\pm ae, 0) = (\pm \sqrt{10}, 0)$	$F_1 = (2, -3) + (\sqrt{10}, 0) = (2 + \sqrt{10}, -3)$ $F_2 = (2, -3) + (-\sqrt{10}, 0) = (2 - \sqrt{10}, -3)$
Vertices	$(\pm a, 0) = (\pm 1, 0)$	$V_1 = (2, -3) + (1, 0) = (3, -3)$ $V_2 = (2, -3) + (-1, 0) = (1, -3)$
Equation of directrices	$X = \pm \frac{a}{e} = \pm \frac{1}{\sqrt{10}}$	$x - 2 = \pm \frac{1}{\sqrt{10}} ; x = 2 + \frac{1}{\sqrt{10}} ; x = 2 - \frac{1}{\sqrt{10}}$

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Top of Form

Ex 5.3

Identify the type of conic section for each of the equations.

Question 1.

$$2x^2 - y^2 = 7$$

Solution:

Comparing this equation with the general equation of the conic

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

We get $A=2, C=-1$

Here $A \neq C$ also A and C are of opposite signs.

So the conic is a hyperbola.

Question 2.

$$3x^2 + 3y^2 - 4x + 3y + 10 = 0$$

Sol. Comparing this equation with the general equation of the conic

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

We get $A=C$ also $B=0$

So the given conic is a circle.

Question 3.

$$3x^2+2y^2=14$$

Solution:

Comparing this equation with the general equation of the conic

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$

We get $A \neq C$ also C are of the same sign.

So the given conic is an ellipse.

Question 4.

$$x^2+y^2+x-y=0$$

Solution:

Comparing this equation with the general equation of the conic

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$

We get $A=C$ and $B=0$

So the given conic is a circle.

Question 5.

$$11x^2-25y^2-44x+50y-256=0$$

Solution:

Comparing this equation with the general equation of the conic

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$

We get $A \neq C$. Also A and C are of opposite sign.

So the conic is a hyperbola.

Question 6.

$$y^2+4x+3y+4=0$$

Solution:

Comparing this equation with the general equation of the conic

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$

We get $A=0$ and $B=0$

So the conic is a parabola.

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Comparing this equation with the general equation of the conic

$$Ax^2+Bxy+Cy^2+Dx+Ey+F=0$$

We get $A=0$ and $B=0$

So the conic is a parabola.

Top of Form

Ex 5.4

Question 1.

Find the equations of the two tangents that can be drawn from (5,2) to the ellipse $2x^2+7y^2=14$.

Solution:

$$2x^2+7y^2=14 \quad (+ \text{ by } 14) \Rightarrow x^2/7 + y^2/2 = 1$$

comparing this equation with $x^2/a^2 + y^2/b^2 = 1$

we get $a^2=7$ and $b^2=2$

The equation of tangent to the above ellipse will be of the form

$$y = mx + \sqrt{a^2m^2 + b^2} \Rightarrow y = mx + \sqrt{7m^2 + 2}$$

Here the tangents are drawn from the point (5,2)

$$\Rightarrow 2 = 5m + \sqrt{7m^2 + 2} \Rightarrow 2 - 5m = \sqrt{7m^2 + 2}$$

Squaring on both sides we get

$$(2 - 5m)^2 = 7m^2 + 2 \Rightarrow 4 - 20m + 25m^2 = 7m^2 + 2 \Rightarrow 18m^2 - 20m + 2 = 0 \quad (+ \text{ by } 2) \Rightarrow 9m^2 - 10m + 1 = 0$$

$$(9m - 1)(m - 1) = 0 \Rightarrow m = 1 \text{ (or) } m = 1/9$$

When $m=1$, the equation of tangent is

$$y = x + 3 \text{ or } x - y + 3 = 0$$

When $m=1/9$ the equation of tangent is

$$y = x/9 + \sqrt{7/81 + 2} \quad (\text{i.e.}) \quad y = x/9 + 13/9 \Rightarrow 9y = x + 13 \text{ or } x - 9y + 13 = 0$$

Question 2.

Find the equations of tangents to the hyperbola $x^2/16 - y^2/64 = 1$ which are parallel to $10x - 3y + 9 = 0$.

Solution:

$$x^2/16 - y^2/64 = 1$$

Here $a^2=16$ and $b^2=64$

The equation of tangents will be of the form $y = mx \pm \sqrt{a^2 m^2 - b^2}$

(i.e.,) $y = mx \pm \sqrt{16m^2 - 64}$

Where 'm' is the slope of the tangent.

Here we are given that the tangents are parallel to $10x - 3y + 9 = 0$

So slope of tangents will be equal to slope of the given line

(i.e.,)

$$m = -(10/-3) = 10/3$$

So equations of tangents will be

$$y = 10/3 x \pm \sqrt{16(10/3)^2 - 64} = 10/3 x \pm \sqrt{16(100/9) - 64} = 10/3 x \pm \sqrt{1600/9 - 576/9} \Rightarrow y = 10/3 x \pm 32/3 \Rightarrow 3y = 10x \pm 32$$

(or) $10x - 3y \pm 32 = 0$.

Question 3.

Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Also find the coordinates of the point of contact.

Solution:

The given ellipse is $x^2 + 3y^2 = 12$

$$(\div \text{ by } 12) \Rightarrow x^2/12 + y^2/4 = 1$$

(i.e.,) Here $a^2 = 12$ and $b^2 = 4$

The given line is $x - y + 4 = 0$

(i.e.,) $y = x + 4$

Comparing this line with $y = mx + c$

We get $m = 1$ and $c = 4$

The condition for the line $y = mx + c$

To be a tangent to the ellipse $x^2/a^2 + y^2/b^2 = 1$ is $c^2 = a^2 m^2 + b^2$

$$\text{LHS} = c^2 = 4^2 = 16$$

$$\text{RHS: } a^2 m^2 + b^2 = 12(1)^2 + 4 = 16$$

LHS = RHS The given line is a tangent to the ellipse. Also the point of contact is

$$(-a^2 m c / b^2 c) = [-(12(1)4)/4, 4] \text{ (i.e.,) } (-3, 1)$$

Question 4.

Find the equation of the tangent to the parabola $y^2 = 16x$ perpendicular to $2x + 2y + 3 = 0$

Solution:

$$y^2 = 16x$$

Comparing this equation with $y^2 = 4ax$

we get $4a = 16 \Rightarrow a = 4$

The equation of tangent to the parabola $y^2=16x$ will be of the form

$$y=mx+am \text{ (Here } a=4) \Rightarrow y=mx+4m$$

Given that the tangent is perpendicular to $2x+2y+3=0$

|| So slope of $2x+2y+3=0$ is $-(2/2)=-1=m_1 \Rightarrow$ slope of its perpendicular $=-1/m_1=-1/-1=1$ (ie.,) $m=1$ ||

So $m=1$

$\Rightarrow$ The equation of tangent will be $y=1(x)+4$ (i.e.) $y=x+4$

(or) $x-y+4=0$

Question 5.

Find the equation of the tangent at $t=2$ to the parabola $y^2=8x$. (Hint: use parametric form)

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Solution:

$$y^2=8x$$

Comparing this equation with $y^2=4ax$ we get $4a=8 \Rightarrow a=2$

Now, the parametric form for $y^2=4ax$ is $x=at^2, y=2at$ Here $a=2$ and $t=2 \Rightarrow x=2(2)^2=8$ and $y=2(2)(2)=8$

So the point is $(8,8)$

Now equation of tangent to $y^2=4ax$ at (x_1, y_1) is $yy_1=2a(x+x_1)$

Here $(x_1, y_1)=(8,8)$ and $a=2$

So equation of tangent is $y(8)=2(2)(x+8)$

(ie.,) $8y=4(x+8)$

($\div$ by 4) $\Rightarrow 2y=x+8 \Rightarrow x-2y+8=0$

Aliter

The equation of tangent to the parabola $y^2=4ax$ at ' t ' is

$$yt=x+at^2$$

Here $t=2$ and $a=2$

So equation of tangent is

(i.e.,) $y(2)=x+2(2)^2$

$2y=x+8 \Rightarrow x-2y+8=0$

Question 6.

Find the equations of the tangent and normal to hyperbola $12x^2-9y^2=108$ at $\theta=\pi/3$.

(Hint: Use parametric form)

Solution:

$$12x^2 - 9y^2 = 108$$

(÷ by 108) we get $12x^2/108 - 9y^2/108 = 108/108 \Rightarrow x^2/9 - y^2/12 = 1$

Comparing this equation with the general form $x^2/a^2 - y^2/b^2 = 1$

we get $a^2 = 9$ and $b^2 = 12$

The parametric form for the hyperbola

$$x^2/a^2 - y^2/b^2 = 1 \text{ is } (a \sec \theta, b \tan \theta)$$

Here $a = 3$ and $b = \sqrt{12} = 2\sqrt{3}$

Also $\theta = \pi/3$

So $a \sec \theta = 3 \times 2 = 6 = x_1$ and $b \tan \theta = 2\sqrt{3}(\sqrt{3}) = 6 = y_1$

So $(x_1, y_1) = (6, 6)$

Now equation of tangent to $x^2/9 - y^2/12 = 1$ at (x_1, y_1) is $xx_1/9 - yy_1/12 = 1$ Here $(x_1, y_1) = (6, 6)$

So equation of tangent is

$$x(6)/9 - y(6)/12 = 1 \Rightarrow 2x/3 - y/2 = 1 \Rightarrow 4x - 3y - 6 = 0$$

Normal is a line perpendicular to tangent

So equation of normal will be of the form $3x + 4y + k = 0$

The normal is drawn at $(6, 6)$

$$\Rightarrow 18 + 24 + k = 0 \Rightarrow k = -42$$

So equation of normal is $3x + 4y - 42 = 0$

Aliter

The equation of tangent to $x^2/a^2 - y^2/b^2 = 1$ at θ is $x \sec \theta - y \tan \theta = 1$

Here $a = 3, b = 2\sqrt{3}$ and $\theta = \pi/3$

So $\sec \theta = \sec \pi/3 = 2$ and $\tan \theta = \tan \pi/3 = \sqrt{3}$

So equation of tangent will be

$$x(2) - y(2\sqrt{3}) = 1 \Rightarrow 2x - 2\sqrt{3}y = 1 \Rightarrow 4x - 3y - 6 = 0$$

(ie.,)

Equation of normal in parametric form is $ax \sec \theta + by \tan \theta = a^2 + b^2$

$$(ie.,) 3(x) + 2\sqrt{3}y(\sqrt{3}) = 9 + 12 \Rightarrow 3x + 4y = 21 \Rightarrow 3x + 4y - 42 = 0$$

Question 7.

Prove that the point of intersection of the tangents at ' t_1 ' and ' t_2 ' on the parabola $y^2 = 4ax$ is $[at_1t_2, a(t_1 + t_2)]$.

Solution:

The equation of tangent to $y^2=4ax$ at ' t ' is given by $yt=x+at^2$

So equation of tangent at ' t_1 ' is $yt_1=x+at_1^2$

and equation of tangent at ' t_2 ' is $yt_2=x+at_2^2$

To find the point of intersection we have to solve the two equations

$$(1) \Rightarrow yt_1 = x + at_1^2$$

$$(2) \Rightarrow yt_2 = x + at_2^2$$

$$(1)-(2) \Rightarrow y(t_1-t_2) = a(t_1^2-t_2^2) \Rightarrow y = a(t_1+t_2)(t_1-t_2)(t_1-t_2) = a(t_1+t_2)y = a(t_1+t_2)$$

Substituting y value in (1) we get

$$a(t_1+t_2)(t_1) = x + at_1^2 \text{ (ie.,)} at_1^2 + at_1t_2 = x + at_1^2 \Rightarrow x = at_1t_2$$

So the point of intersection is $[at_1t_2, a(t_1+t_2)]$

Question 8.

If the normal at the point ' t_1 ' on the parabola $y^2=4ax$ meets the parabola again at the point ' t_2 ', then prove that $t_2 = -(t_1+2t_1)$

Solution:

Equation of normal to $y^2=4ax$ at ' t ' is $y+xt=2at+at^3$.

So equation of normal at ' t_1 ' is $y+xt_1=2at_1+at_1^3$

The normal meets the parabola $y^2=4ax$ at ' t_2 ' (ie.,) at $(at_2^2, 2at_2)$

$$\Rightarrow 2at_2 + at_1t_2^2 = 2at_1 + at_1^3 \text{ So}$$

$$2a(t_2-t_1) = at_1^3 - at_1t_2^2 \Rightarrow 2a(t_2-t_1) = at_1(t_1^2-t_2^2) \Rightarrow 2(t_2-t_1) = t_1(t_1+t_2)$$

$$(t_1-t_2) \Rightarrow 2 = -t_1(t_1+t_2) \Rightarrow t_1+t_2 = -2t_1 \Rightarrow t_2 = -t_1-2t_1 = -(t_1+2t_1)$$

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Additional Questions

Question 1.

Find the equations of the tangent and normal to the parabolas : $x^2+2x-4y+4=0$ at $(0,1)$

Solution:

Equation of the tangent at (x_1, y_1) is

$$xx_1 + 2(x+x_1) - 4(y+y_1) + 4 = 0; \text{ Here, } (x_1, y_1) = (0, 1)$$

$\therefore$ Equation of the tangent is

$$x(0) + 2(x+0) - 4(y+1) + 4 = 0; \text{ i.e., } x - 4(y+1) + 4 = 0 \Rightarrow 2x - 4y - 4 + 4 = 0 \Rightarrow 2x - 4y + 4 = 0 \text{ (} \div \text{ by 2)} \Rightarrow x - 2y + 2 = 0$$

Slope of the tangent $= -(1-2) = 1 = m$. $\therefore$ Slope of the normal

$$= -1/m = -2 \text{ (at } (x_1, y_1) = (0, 1))$$

Equation of the normal $y-1=-2(x-0) \Rightarrow y-1=-2x$

$$\therefore 2x+y-1=0$$

Question 2.

Find the equations of the tangent and normal to the parabola $y^2=8x$ at $t=12$

Solution:

Equation of the parabola is $y^2=8x$. Here $4a=8 \Rightarrow a=2$

't' represents the point $(at^2, 2at)=(2t^2, 4t)$; Here, $t=12$

$\therefore$ The point is $2(14), 4(12)=(12, 2)$; i.e., $(x_1, y_1)=(12, 2)$

So equation of the tangent to $y^2=8x$ at $(12, 2)$ is $y(y_1)=8(x+x_1)$

i.e., $y(2)=8(x+12) \Rightarrow 4y=8(2x+1) \Rightarrow 2y=4x+8 \Rightarrow 4x-2y+8=0$ ($\div$ by 2) $\Rightarrow 2x-y+4=0$ Slope of the tangent $=-(2-1)=-1=m$. $\therefore$ Slope of the normal $=-1/m=1$

Equation of the normal $y-2=1(x-12)$

$$2(y-2)=x-12 \Rightarrow 2y-4=x-12 \Rightarrow x-2y+8=0$$

$\therefore$ Equation of the tangent is $2x-y+4=0$ and equation of the normal is $x-2y+8=0$

Question 3.

Find the equations of the tangents: to the parabola $y^2=6x$, parallel to $3x-2y+5=0$

Solution:

$$y^2=6x; 4a=6 \Rightarrow a=1.5$$

Equation of the tangent is $y=mx+a/m$

Here, the tangent is parallel to $3x-2y+5=0$.

$\therefore$ Slope of the tangent = slope of the line $=-(3-2)=-1/2=m$ and $a=3/2$

So equation of the tangent is $y=3/2x+3/2$

$$y=3/2x+3/2 \Rightarrow 2y=3x+3$$

$\therefore$ Equation of the tangent is $3x-2y+3=0$

Question 4.

Find the equations of the tangents: to the parabola $4x^2-y^2=64$ Which are parallel to $10x-3y+9=0$.

Solution:

Slope of $10x-3y+9=0$ is $-(10-3)=-7/3=m$

$\therefore$ Slope of its parallel $=m=-7/3$

The given hyperbola is $4x^2-y^2=64$.

$$\text{i.e., } 4x^2 - y^2 = 1 \Rightarrow x^2 - \frac{y^2}{4} = \frac{1}{4}$$

Comparing this with $x^2 - \frac{y^2}{b^2} = 1$ we get, $a^2 = 1, b^2 = 4$

Here, $m = 103$

So the equation of the tangent is $y = mx \pm \sqrt{a^2 m^2 - b^2}$

$$\text{i.e., } y = 103x \pm \sqrt{16(103)^2 - 64} \text{ i.e.,}$$

$$y = 103x \pm \sqrt{16(1009) - 64} = 103x \pm \sqrt{1600 - 576} = 103x \pm \sqrt{1024} = 103x \pm 32$$

Question 5.

Find the equation of the tangent from point $(2, -3)$ to the parabola $y^2 = 4x$.

Solution:

$$y^2 = 4x$$

Equation of the tangent to the parabola will be of the form $y = mx + \frac{1}{m}$. The tangents pass through $(2, -3)$

$$\Rightarrow -3 = 2m + \frac{1}{m} \Rightarrow -3m = 2m^2 + 1 \Rightarrow 2m^2 + 3m + 1 = 0$$

$$m = \frac{-3 \pm \sqrt{9 - 8}}{4} = \frac{-3 \pm 1}{4} \Rightarrow m = -1, -\frac{1}{2}$$

When $m = -1$, the equation of the tangent is

$$y = -x + 1; \text{ i.e., } x + y - 1 = 0$$

When $m = -\frac{1}{2}$, equation of the tangent is

$$y = -\frac{1}{2}x + 2 \Rightarrow x + 2y - 4 = 0$$

The two tangents drawn from $(2, -3)$ are $x + y - 1 = 0, x + 2y - 4 = 0$

Question 6.

Find the equation of the tangents from the point $(2, -3)$ to the parabola $2x^2 - 3y^2 = 6$

Solution:

$$2x^2 - 3y^2 = 6$$

$$(\div \text{ by } 6) \Rightarrow \frac{2x^2}{6} - \frac{3y^2}{6} = 1; \text{ i.e., } \frac{x^2}{3} - \frac{y^2}{2} = 1; a^2 = 3; b^2 = 2$$

Equation of the tangent is of the form $y = mx \pm \sqrt{a^2 m^2 - b^2}$

$$\text{i.e., } y = mx \pm \sqrt{3m^2 - 2}$$

The tangents are drawn from $(2, -3) \Rightarrow -3 = 2m \pm \sqrt{3m^2 - 2}$

$$\text{i.e., } -(3 + 2m) = \pm \sqrt{3m^2 - 2}$$

Squaring on both sides, $(3 + 2m)^2 = 3m^2 - 2$

$$m^2 + 2m + 1 - 3m^2 + 2 = 0 \Rightarrow -2m^2 + 2m + 3 = 0 \text{ i.e.,}$$

$$2m^2 - 2m - 3 = 0 \Rightarrow m = \frac{2 \pm \sqrt{4 + 24}}{4} = \frac{2 \pm \sqrt{28}}{4} = \frac{1 \pm \sqrt{7}}{2} \Rightarrow m_1 = \frac{1 + \sqrt{7}}{2}, m_2 = \frac{1 - \sqrt{7}}{2}$$

When $m_1 = 1 + \sqrt{72}$, equation of the the tangent is

$$y = (1 + \sqrt{72})x + \frac{1}{1 + \sqrt{72}}$$

$$\sqrt{3(1 + \sqrt{72})^2 - 2y} = (1 + \sqrt{72})x + \sqrt{3 + 6\sqrt{7} + 21} - 822y = (1 + \sqrt{72})x + \sqrt{16 + 6\sqrt{7}}$$

Similarly by substituting $m = 1 - \sqrt{72}$, equation of the the tangent is

$$2y = (1 - \sqrt{72})x + \sqrt{16 - 6\sqrt{7}}$$

Question 7.

Prove that the line $5x + 12y = 9$ touches the hyperbola $x^2 - 9y^2 = 9$ and find the point of contact.

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Solution:

The given line is $5x + 12y = 9 \Rightarrow y = -\frac{5}{12}x + \frac{3}{4}$

Comparing the above line with $y = mx + c$ we get,

$$m = -\frac{5}{12} \text{ and } c = \frac{3}{4}$$

The given hyperbola is $x^2 - 9y^2 = 9$

i.e., $\frac{x^2}{9} - \frac{y^2}{1} = 1$; Here, $a^2 = 9; b^2 = 1$

Condition for the line $y = mx + c$ to be a tangent to the hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \text{ is } c^2 = a^2m^2 - b^2.$$

$$\text{Here, } c^2 = \left(\frac{3}{4}\right)^2 = \frac{9}{16}$$

$$a^2m^2 - b^2 = 9\left(-\frac{5}{12}\right)^2 - 1 = 9\left(\frac{25}{144}\right) - 1 = \frac{25}{16} - 1 = \frac{9}{16}$$

(1) = (2) $\Rightarrow$ The given line is a tangent to the curve i.e., the given line touches the curve.

To find the point of contact we have to solve the line and the curve.

The given line $5x + 12y = 9$

$$\Rightarrow 12y = 9 - 5x \Rightarrow y = \frac{9 - 5x}{12}$$

Substituting $y = \frac{9 - 5x}{12}$ in $x^2 - 9y^2 = 9$ we get,

$$x^2 - 9\left[\frac{9 - 5x}{12}\right]^2 = 9 \Rightarrow x^2 - \frac{9(81 - 90x + 25x^2)}{144} = 9 \Rightarrow x^2 - \frac{81 + 25x^2 - 90x}{16} = 9 \Rightarrow 16x^2 - (81 + 25x^2 - 90x) = 144 \Rightarrow 16x^2 - 81 - 25x^2 + 90x = 144$$

$$\Rightarrow -9x^2 + 90x - 225 = 0 \Rightarrow (x - 5)^2 = 0 \Rightarrow x = 5 \text{ when } x = 5, y = \frac{9 - 25}{12} = -\frac{16}{12} = -\frac{4}{3}$$

Point of contact is $(5, -\frac{4}{3})$.

Question 8.

Show that the line $x - y + 4 = 0$ is a tangent to the ellipse $x^2 + 3y^2 = 12$. Find the co-ordinates of the point of contact.

Solution:

The given ellipse is $x^2 + 3y^2 = 12$

Given line is $x - y + 4 = 0 \Rightarrow y = x + 4$

Here, $m = 1$ and $c = 4$

The condition for the line $y = mx + c$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is $c^2 = a^2 m^2 + b^2$.

LHS: $c^2 = 4^2 = 16$

RHS: $a^2 m^2 + b^2 = 12(1) + 4 = 16$

LHS: RHS $\Rightarrow$ the given line is a tangent to the ellipse.

Ex 5.5

Question 1.

A bridge has a parabolic arch that is 10 m high in the centre and 30 m wide at the bottom. Find the height of the arch 6 m from the centre, on either sides.

Solution:

From the diagram, equation of the parabolic arch

$$x^2 = -4ay$$

Use the point (15, -10) in (1)

$$(1) \Rightarrow 15^2 = -4a(-10) \quad 225 = 40a \quad \text{substitute } a = \frac{225}{40} \text{ in (1)} \quad (1) \Rightarrow x^2 = -4\left(\frac{225}{40}\right)y$$

Use the point (6, -y₁) in (2)

$$(2) \Rightarrow (6)^2 = -4\left(\frac{225}{40}\right)(-y_1) \quad 36 \times 10 \times 225 = y_1$$

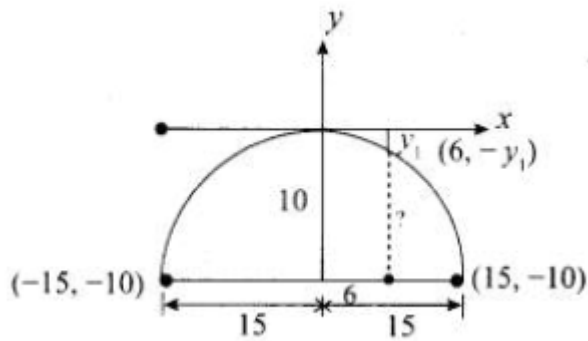
$\therefore$ The required height $= 10 - y_1 = 10 - 1.6 = 8.4$ m.

Question 2.

A tunnel through a mountain for a four lane highway is to have a elliptical opening. The total width of the highway (not the opening) is to be 16 m, and the height at the edge of the road must be sufficient for a truck 4 m high to clear if the highest point of the opening is to be 5 m approximately. How wide must the opening be?

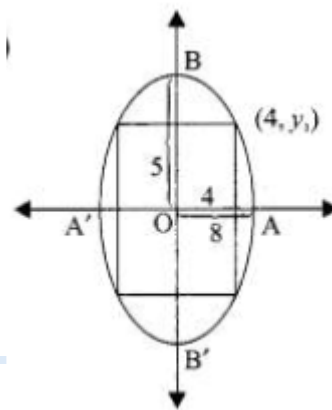
Solution:

From the diagram,



$AA' = 16$ m, $OA = 8$ m, $OB = 5$ m

$\therefore$ Equation of the ellipse is



$$\frac{x^2}{25} + \frac{y^2}{16} = 1$$

$\therefore$ The point $(4, y_1)$ on the ellipse

$$(1) \Rightarrow \frac{4^2}{25} + \frac{y_1^2}{16} = 1 \Rightarrow \frac{y_1^2}{16} = 1 - \frac{16}{25} \Rightarrow y_1 = \sqrt{9 \times \frac{16}{25}} = 3 \times \frac{4}{5} = 4.8 \text{ m}$$

$\therefore$ The required width for the opening is $2y_1 = 2(4.8) = 9.6$ m

Question 3.

At a water fountain, water attains a maximum height of 4 m at horizontal distance of 0.5 m from its origin. If the path of water is a parabola, find the height of water at a horizontal distance of 0.75 m from the point of origin.

Solution:

From the diagram

Equation of the path of water

$$x^2 = -4ay$$

Use the point $(0.5, -4)$ in (1)

$$(0.5)^2 = -4a(-4) \Rightarrow (0.5)^2 = 16a \Rightarrow a = 0.025 \text{ substitute in (1)} \Rightarrow x^2 = -4[(0.025)16]y$$

Use the point $(0.75, -y_1)$ in (2)

$$(2) \Rightarrow (0.25)^2 = -4[(0.25)16](-y_1) \Rightarrow (0.25)^2 = 0.254y_1 \Rightarrow y_1 = 1 \text{ m}$$

The refined height $= 4 - y_1 = 4 - 1 = 3 \text{ m}$

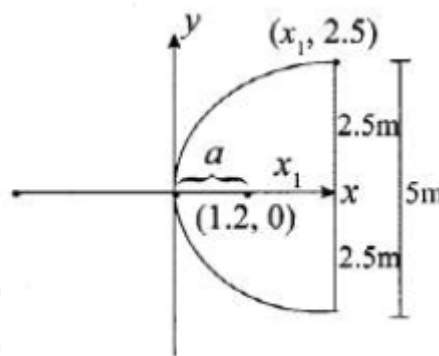
Question 4.

An engineer designs a satellite dish with a parabolic cross section. The dish is 5 m wide at the opening, and the focus is placed 1.2 m from the vertex

- Position a coordinate system with the origin at the vertex and the x-axis on the parabola's axis of symmetry and find an equation of the parabola.
- Find the depth of the satellite dish at the vertex.

Solution:

From the diagram,



- Consider the satellite dish is open rightward parabola $y^2 = 4ax$ (1)

Clearly $a = 1.2 \text{ m}$

$$(1) \Rightarrow y^2 = 4(1.2)x \Rightarrow y^2 = 4.8x$$

- Use the point $(x_1, 2.5)$ in (1)

$$(2.5)^2 = 4(1.2)x_1$$

$$(2.5)^2 4(1.2) = y_1$$

$$x_1 = 1.3 \text{ m}$$

$\therefore$ The depth of the satellite dish at vertex is 1.3 m

Question 5.

Parabolic cable of a 60 m portion of the roadbed of a suspension bridge are positioned as shown below. Vertical Cables are to be spaced every 6 m along this portion of the roadbed. Calculate the lengths of first two of these vertical cables from the vertex.

Solution:

From the diagram,

Equation of the suspension bridge

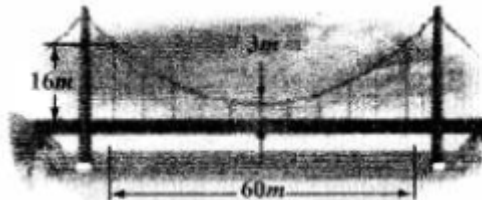
$$(x-h)^2=4a(y-k)$$

But V (0,3)

$$x^2=4a(y-3)$$

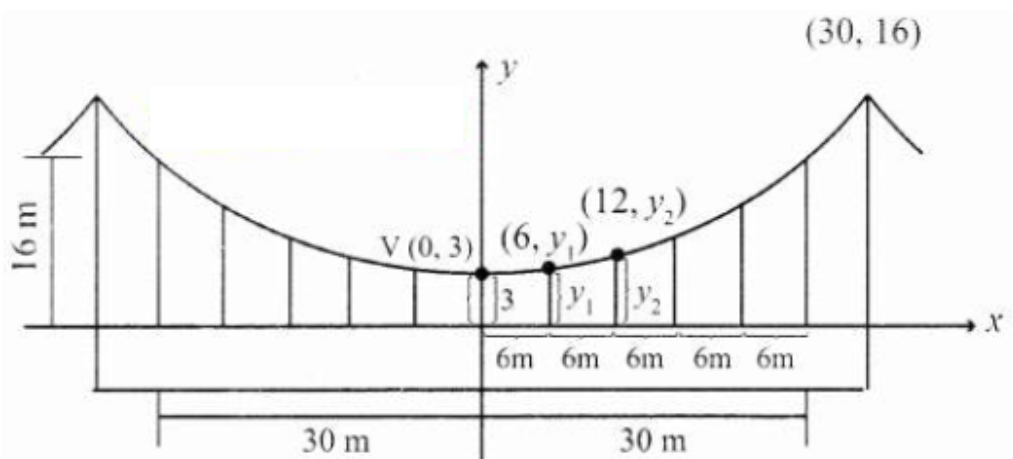
Use the point (30,16) in (1)

$$30^2=4a(16-3) \Rightarrow 900=13 \times 4a \Rightarrow 900/13 \times 4 = a$$



$\Rightarrow$

$$x^2=4\left(\frac{900}{13} \times 4\right)(y-3) \Rightarrow x^2=\left(\frac{900}{13}\right)(y-3)$$



(i) The length of the first vertical cable from the vertex is

Use (6, y_1) in (2)

$$(2) \Rightarrow (6)^2 = \frac{900}{13}(y_1 - 3) \Rightarrow 36 \times \frac{13}{900} = y_1 - 3 \Rightarrow y_1 - 3 = 0.52 \Rightarrow y_1 = 3.52 \text{ m}$$

(ii) The length of the second vertical cable from the vertex is

Use the point (12, y_2) in (2)

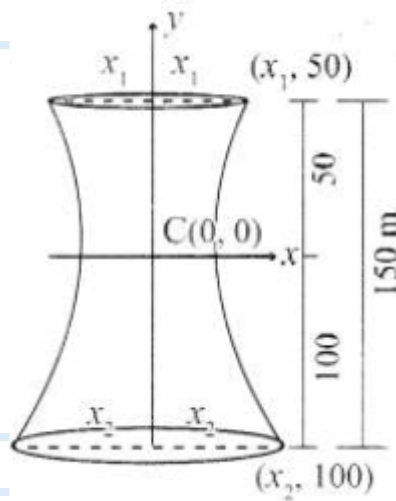
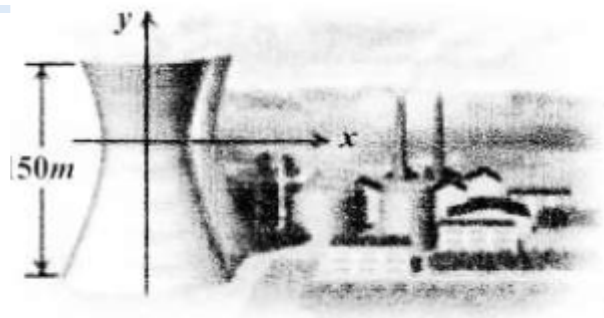
$$(2) \Rightarrow (12)^2 = \frac{900}{13}(y_2 - 3) \Rightarrow 144 \times \frac{13}{900} = y_2 - 3 \Rightarrow y_2 - 3 = 0.52 \Rightarrow y_2 = 3.52 \text{ m}$$

Question 6.

Cross section of a Nuclear cooling tower is in the shape of a hyperbola with equation $\frac{x^2}{30^2} - \frac{y^2}{44^2} = 1$. The tower is 150 m tall and the distance from the top of the tower to the centre of the hyperbola is half the distance from the base of the tower to the centre of the hyperbola. Find the diameter of the top and base of the tower.

Solution:

From the diagram, equation of hyperbola is



Substitute $(x_1, 50)$ in (1)

$$x_1^2(30)^2 - (50)^2(44)^2 = 1 \quad x_1^2(30)^2 = 1 + (50)^2(44)^2 \quad x_1^2 = \frac{1 + (50)^2(44)^2}{(30)^2} \quad x_1 = \frac{\sqrt{1 + (50)^2(44)^2}}{(30)} \quad x_1 = 45.41 \text{ m diameter}$$

Sub $(x_2, 100)$ in (1)

$$x_2^2(30)^2 - (100)^2(44)^2 = 1 \quad x_2^2 = \frac{1 + (100)^2(44)^2}{(30)^2} \quad x_2 = \frac{\sqrt{1 + (100)^2(44)^2}}{(30)} \quad x_2 = 74.49 \text{ m diameter}$$

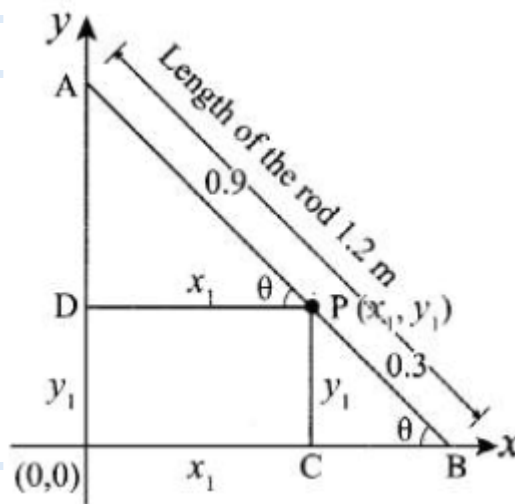
Question 7.

A rod of length 1.2 m moves with its ends always touching the coordinate axes. The locus of a point P on the rod, which is 0.3 m from the end in contact with x-axis is an ellipse. Find the eccentricity.

Solution:

From the diagram,

(i) $\triangle OAB$ be a right angle triangle.



(ii) $\angle APD$ and $\angle PBC$ are corresponding angles, so corresponding angles are equal.

In $\triangle ADP$ In $\triangle PCB$ $\cos \theta = \frac{\text{adj}}{\text{hyp}} \sin \theta = \frac{\text{opp}}{\text{hyp}} \cos \theta = \frac{x_1}{0.9} \sin \theta = \frac{y_1}{0.3}$

We know that

$$\cos^2 \theta + \sin^2 \theta = 1 \Rightarrow \left(\frac{x_1}{0.9}\right)^2 + \left(\frac{y_1}{0.3}\right)^2 = 1$$

$\therefore$ The locus of $P(x_1, y_1)$ is an ellipse.

$$x^2(0.9)^2 + y^2(0.3)^2 = 1 \Rightarrow a^2 = (0.9)^2, b^2 = (0.3)^2 \text{ Eccentricity}$$

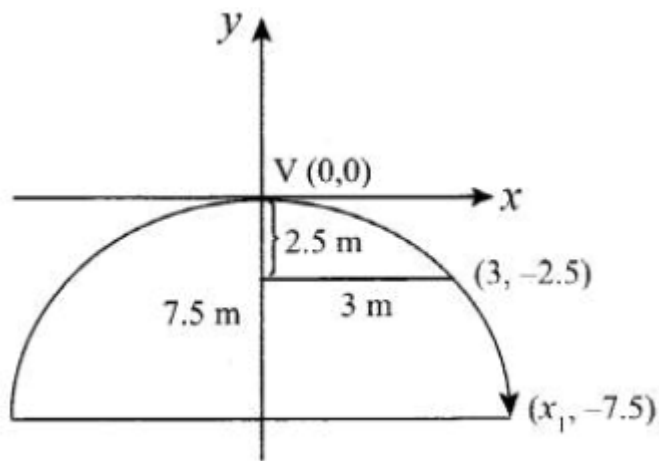
$$= \sqrt{a^2 - b^2} = \sqrt{(0.9)^2 - (0.3)^2} = \sqrt{0.81 - 0.09} = \sqrt{0.72} = \sqrt{\frac{72}{100}} = \frac{\sqrt{72}}{10} = \frac{6\sqrt{2}}{10} = \frac{3\sqrt{2}}{5}$$

Question 8.

Assume that water issuing from the end of a horizontal pipe, 7.5 m above the ground, describes a parabolic path. The vertex of the parabolic path is at the end of the pipe. At a position 2.5 m below the line of the pipe, the flow of water has curved outward 3 m beyond the vertical line through the end of the pipe. How far beyond this vertical line will the water strike the ground?

Solution:

From the diagram,



Equation of the water path is $x^2 = -4ay$

Use the point $(3, -2.5)$ in (1)

$$(3)^2 = -4a(-2.5)$$

$$9 = 10a$$

$a = 9/10$ substituting in (1)

$$(1) \Rightarrow x^2 = -4(9/10)y$$

Use the point $(x_1, -7.5)$ in (2)

$$(2) \Rightarrow x_1^2 = -4(9/10)(-7.5) \Rightarrow x_1^2 = 30(9/10) \Rightarrow x_1 = \sqrt{3 \times 9} = 3\sqrt{3} \text{ m}$$

$\therefore$ The water strikes the ground $3\sqrt{3}$ m beyond the vertical line.

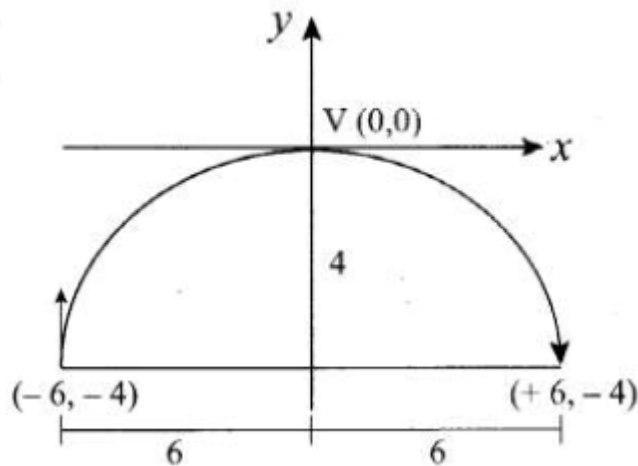
Question 9.

On lighting a rocket cracker it gets projected in a parabolic path and reaches a maximum height of 4 m when it is 6 m away from the point of projection. Finally it reaches the ground 12 m away from the starting point.

Find the angle of projection?

Solution:

From the diagram,



Equation of the parabolic path is

$$x^2 = -4ay$$

Use the point (6, -4) in (1)

$$(1) \Rightarrow (6)^2 = 16a$$

$$36 = 16a \text{ substitute } a = \frac{9}{4} \text{ in (1)} \Rightarrow x^2 = -4\left(\frac{9}{4}\right)y \Rightarrow x^2 = -9y \dots \dots \dots (2)$$

Differentiate with respect to 'x'

$$2x = -9 \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = -\frac{2x}{9}$$

$\therefore$ The slope at (-6, -4)

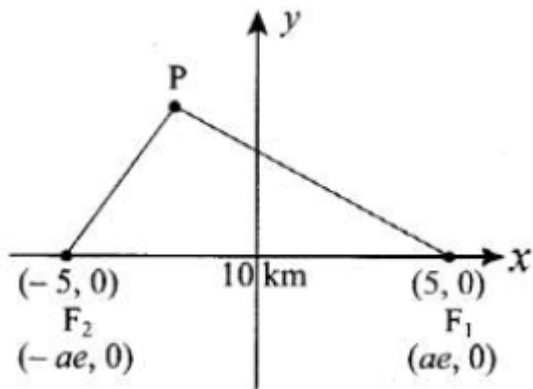
$$\tan \theta = -\frac{2(-6)}{9} \Rightarrow \tan \theta = \frac{4}{3} \Rightarrow \theta = \tan^{-1}\left(\frac{4}{3}\right)$$

Question 10.

Points A and B are 10 km apart and it is determined from the sound of an explosion heard at those points at different times that the location of the explosion is 6 km closer to A than B. Show that the location of the explosion is restricted to a particular curve and find an equation of it.

Solution:

From the diagram,



$$F_1P \sim F_2P = 6a = 3ae = 5 \Rightarrow e = \frac{5}{3} \quad b^2 = a^2[e^2 - 1] = 9[25 - 9] \quad b^2 = 16$$

$\therefore$ The required equation is

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{9} - \frac{y^2}{16} = 1$$

Top of Form

Additional Problems

Question 1.

If a parabolic reflector is 20 cm in diameter and 5 cm deep, find the distance of the focus from the centre of the reflector.

Solution:

By the property of the parabolic reflector, the position of the bulb should be placed at the focus.

By taking the vertex at the origin the equation of the reflector is $y^2 = 4ax$.

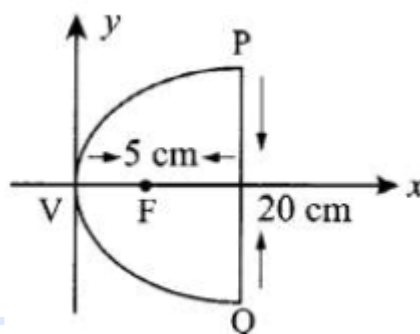
Let PQ be the diameter of the reflector $P = (5, 10)$

Since $P(5, 10)$ lies on the parabola,

$$10^2 = 4a \times 5$$

$$\text{i.e., } 100 = 20a \Rightarrow a = 5$$

So the focus is at a distance of 5 cm from the vertex and focus is $(5, 0)$.



Question 2.

The focus of a parabolic mirror is at a distance of 8 cm from its centre (vertex). If the mirror 25 cm deep, find the diameter of the mirror.

Solution:

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Let the vertex be at the origin.

$$VF = a = 8 \text{ cm}$$

The equation of the parabola is

$$y^2 = 4ax = 4(8)x = 32x$$

Depth of the mirror $= x = 25 \text{ cm}$

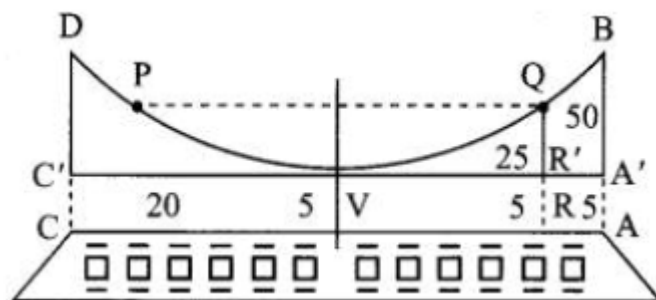
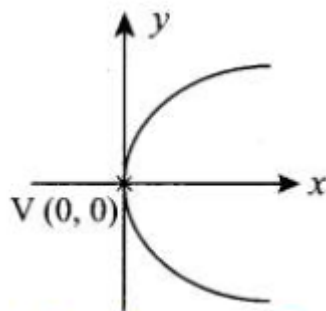
So, radius is y .

$\Rightarrow y^2 = 32(25) = 800 \Rightarrow y = \sqrt{800} = 10\sqrt{8} = 10 \times 2\sqrt{2} = 20\sqrt{2}$ = Radius of the mirror $\therefore$ Diameter of the mirror $= 2 \times 20\sqrt{2} = 40\sqrt{2} \text{ cm}$ of the mirror.

Question 3.

A cable of a suspension bridge is in the form of a parabola whose span is 40mts. The roadway is 5mts below the lowest point of the cable. If an extra support is provided across the cable 30mts above the ground level, find the length of the support if the height of the pillars are 55mts.

Solution:



The lowest point on the cable is taken as the vertex and it is taken as the origin. Let AB, CD be the pillars.

Span of parabola = 40mts = distance between AB and CD

$$C'V = VA' = 20\text{mts}$$

Height of each pillar = 55mts $\Rightarrow AB = 55\text{mts}$

$$\text{So, } A'B = 55 - 5 = 50\text{mts}$$

Thus, the point B is (20, 50).

Equation of the parabola is $x^2 = 4ay$

Here, B is a point on the parabola, $x^2 = 4ay$

$$(20)^2 = 4a(50) \Rightarrow 4a = 20 \times 20 / 50 = 8$$

$\therefore$ The equation is $x^2 = 8y$

Let PQ be the length of the extra support RQ.

$$RQ = 30, R' = 5 \Rightarrow R'Q = 25$$

Let VR' be x_1 . Q is $(x_1, 25)$.

Q is a point on parabola

$$x_1^2 = 8 \times 25 = 200 \Rightarrow x_1 = \sqrt{200} = 10\sqrt{2}$$

The entire length, $PQ = 2x_1 = 20\sqrt{2}\text{mts}$.

Question 4.

A kho-kho player in a practice session while running realises that the sum of the distances from the kho-kho poles from him is always 8 m. Find the equation of the path traced by him if the distance between the poles is 6 m

Solution:

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$$\text{Give } FP + F'P = 4$$

$$\text{ie., } 2a = 8 \text{ and}$$

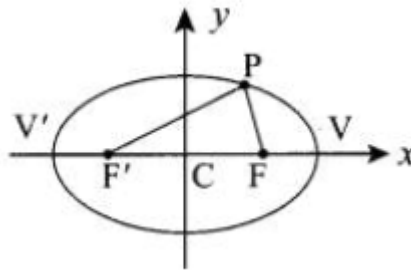
$$FF' = 2ae = 6$$

$$\text{ie., } a = 4 \text{ and } ae = 3$$

$$e = aea = 3/4$$

$$b^2 = a^2(1 - e^2) = 16(1 - 9/16) = 7$$

So the equation of the path is an ellipse whose equation is $x^2/16 + y^2/7 = 1$.



Ex 5.6

Choose the correct or the most suitable answer from the given four alternatives.

Question 1.

The equation of the circle passing through (1,5) and (4,1) and touching y-axis is $x^2+y^2-5x-6y+9+\lambda(4x+3y-19)=0$ where λ is equal to

- (a) 0,-409
- (b) 0
- (c) 409
- (d) -409

Solution:

- (a) 0,-409

Hint:

$$x^2+y^2-5x-6y+9+\lambda(4x+3y-19)=0 \text{ ie., } x^2+y^2+x(4\lambda-5)+y(3\lambda-6)+9-19\lambda=0$$

Since it touches y axis $x=0$

$$\Rightarrow y^2+y(3\lambda-6)+9-19\lambda=0$$

It is a quadratic in y,

Since the circle touches y axis the roots must be equal $\Rightarrow b^2-4ac=0$

$$\text{ie., } (3\lambda-6)^2-4(1)(9-19\lambda)=0$$

Solving we get $\lambda=0$ or -409

Question 2.

The eccentricity of the hyperbola whose latus rectum is 8 and conjugate axis is equal to half the distance between the foci is.

- (a) 43
- (b) $4\sqrt{3}$
- (c) $2\sqrt{3}$
- (d) 32

Solution:

(c) $2\sqrt{3}$

Hint:

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Given $2b^2a=8$ and $2b=ae$

$\Rightarrow b=ae^2 \Rightarrow b^2=a^2e^4$ but

$b^2=a^2(e^2-1)=a^2e^2-a^2a^2e^2-a^2=a^2e^4 \Rightarrow a^2e^2-a^2e^4=a^2a^2e^4=a^2e=\sqrt{43}=2\sqrt{3}$

Question 3.

The circle $x^2+y^2=4x+8y+5$ intersects the line $3x-4y=m$ at two distinct points if

- (a) $15 < m < 65$
- (b) $35 < m < 85$
- (c) $-85 < m < -35$
- (d) $-35 < m < 15$

Solution:

(d) $-35 < m < 15$

Hint:

$$x^2+y^2-4x-8y-5=0$$

$3x-4y=m$ Solving (1) and (2) we get

from (2) $\Rightarrow 3x=4y+m$

$$x=\frac{4y+m}{3}$$

Substituting x value in (1) we get

$$(4y+m)^2+y^2-4(4y+m)-8y-5=0$$

It is a quadratic in y and given that the roots are distinct

$$\Rightarrow b^2-4ac > 0$$

On simplifying we get

$$\Rightarrow -9m^2-18m+4725 > 0 \Rightarrow m^2+20m-525 < 0 \Rightarrow (m+35)(m-15) \leq 0 \Rightarrow m \text{ lies between } -35 \text{ and } 15 \text{ ie, } -35 < m < 15$$

Question 4.

The length of the diameter of the circle which touches the x-axis at the point (1,0) and passes through the point (2,3).....

- (a) 65
- (b) 53

(c) 103

(d) 35

Solution:

(c) 103

Hint:

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Since radius is $\perp$ to tangent it passes through the centre (1,y)

AC=CB= radius

$$AC^2=CB^2 \Rightarrow y^2=1+(y-3)^2$$

$$6y=10 \Rightarrow y=5/3$$

Diameter $2y=10/3$

Question 5.

The radius of the circle $3x^2+by^2+4bx-6by+b^2=0$ is

(a) 1

(b) 3

(c) $\sqrt{10}$

(d) $\sqrt{11}$

Solution:

(c) $\sqrt{10}$

Hint:

Here $b=3$

$$3x^2+3y^2+12x-18y+9=0 \Rightarrow x^2+y^2+4x-6y+3=0$$

Comparing with general form

$$g=2, f=-3, c=3. \therefore \text{radius} = \sqrt{g^2+f^2-c} = \sqrt{4+9-3} = \sqrt{10}$$

Question 6.

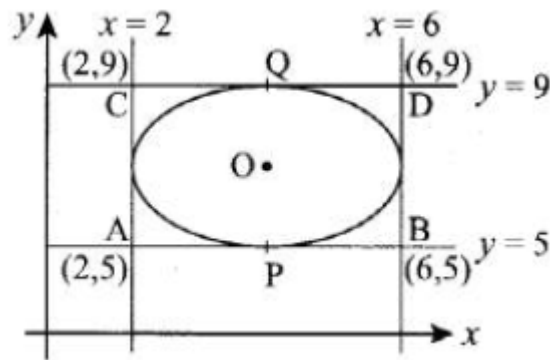
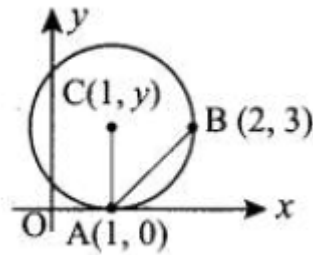
The centre of the circle inscribed in a square formed by the lines $x^2-8x-12=0$ and $y^2-14y+45=0$ is

(a) (4,7)

(b) (7,4)

(c) (9,4)

(d) (4,9)



Solution:

(a) (4,7)

Hint:

$$(x-6)(x-2)=0 \Rightarrow x=2 \text{ or } 6 \quad (y-9)(y-5)=0 \Rightarrow y=5 \text{ or } 9$$

$$P = \text{mid point of } AB = \left(\frac{2+6}{2}, \frac{5+5}{2}\right) = (4, 5) \quad Q = \text{mid point of } CD = \left(\frac{2+6}{2}, \frac{9+9}{2}\right) = (4, 9)$$

Centre of the circle = Mid point of PQ

$$= \left(\frac{4+4}{2}, \frac{5+9}{2}\right) = (4, 7)$$

Question 7.

The equation of the normal to the circle $x^2 + y^2 - 2x - 2y + 1 = 0$ which is parallel to the line $2x + 4y = 3$ is .

- (a) $x + 2y = 3$
- (b) $x + 2y + 3 = 3$
- (c) $2x + 4y + 3 = 0$
- (d) $x - 2y + 3 = 0$

Solution:

(a) $x + 2y = 3$

Hint:

Centre of the circle = (1,1) and radius = $r=1$

Normal is parallel to $2x + 4y = 3$

So equation of normal will be the form $2x+4y=k$

It passes through the centre $(1,1) \Rightarrow 2+4=6=k$

So equation of the normal is $x+4y=6$

(+ by 2) $x+2y=3$

Question 8.

If $P(x,y)$ be any point on $16x^2+25y^2=400$ with foci $F_1(3,0)$ and $F_2(-3,0)$ then PF_1+PF_2 is

(a) 8

(b) 6

(c) 10

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(d) 12

Solution:

(c) 10

Hint:

$F_1P+F_2P=2a$

Here the equation is $16x^2+25y^2=400$

(+by 400) $\Rightarrow 16x^2/400+25y^2/400=1 \Rightarrow x^2/25+y^2/16=1$ $a^2=25 \Rightarrow a=5 \therefore 2a=10$

Question 9.

The radius of the circle passing through the point $(6,2)$ two of whose diameter are $x+y=6$ and $x+2y=4$ is

(a) 10

(b) $2\sqrt{5}$

(c) 6

(d) 4

Solution:

(b) $2\sqrt{5}$

Hint:

The point of intersection of the diameters is the centre.

Now solving $x+y=6$.

$x+2y=4$ (2)

(1) -(2) $\Rightarrow -y=2 \Rightarrow y=-2$

Substituting $y=-6$ in (1)

$x-2=6 \Rightarrow x=8$ Centre $= (8, -2)$ The circle passes through $(6, 2) \therefore$ radius $= \sqrt{(8-6)^2 +$

$$(-2-2)^2 = \sqrt{4+16} = \sqrt{20} = \sqrt{4 \times 5} = 2\sqrt{5}$$

Question 10.

The area of quadrilateral formed with foci of the hyperbolas $x^2/a^2 - y^2/b^2 = 1$ and $x^2/a^2 - y^2/b^2 = -1$ is

(a) $4(a^2 + b^2)$

(b) $2(a^2 + b^2)$

(c) $a^2 + b^2$

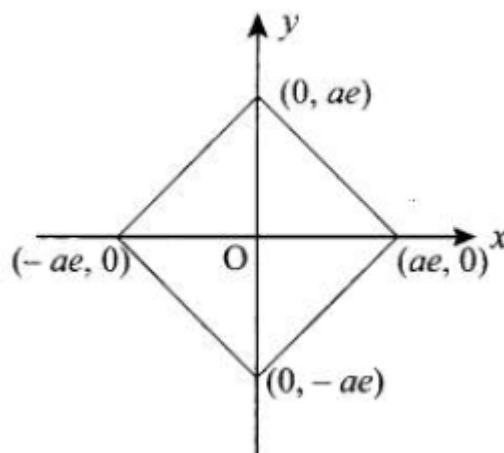
(d) $12(a^2 + b^2)$

Solution:

(b) $2(a^2 + b^2)$

Hint:

The foci are $(\pm ae, 0)$ and $(0, \pm ae)$



The above diagram is a rhombus.

Its area $= \frac{1}{2} d_1 d_2$

$$= \frac{1}{2} (2ae) (2ae) = 2a^2e^2$$

But we know $b^2 = a^2(e^2 - 1)$

$$\text{i.e., } b^2 = a^2e^2 - a^2 \Rightarrow b^2 + a^2 = a^2e^2$$

So area $= 2(a^2 + b^2)$

Question 11.

If the normals of the parabola $y^2 = 4x$ drawn at the end points of its latus rectum are tangents to the circle $(x-3)^2 + (y+2)^2 = r^2$, then the value of r^2 is

(a) 2

(b) 3

(c) 1

(d) 4

Solution:

(a) 2

Hint:

The normals are $x \pm y = 3$

Distance from $(3, -2)$ on both normal is r

(i.e.) $r = \frac{|3 - 2 - 3\sqrt{2}|}{\sqrt{2}} = \sqrt{2} \Rightarrow r^2 = 2$

Question 12.

If $x + y = k$ is a normal to the parabola $y^2 = 12x$, then the value of k is

(a) 3

(b) -1

(c) 1

(d) 9

Solution:

(d) 9

Hint:

Slope of normal $= -1$ and Slope of tangent $= 1$

The condition for $y = mx + c$ to be a tangent to the parabola

$y^2 = 4ax$ is $c = am$

Here the parabola is $y^2 = 12x \Rightarrow a = 3$ and $m = 1$

$\therefore c = 3 \cdot 1 = 3$

$\therefore$ Equation of tangent is $y = x + 3$

Solving the parabola and tangent we get the points $(3, 6)$ and $(3, -6) \therefore k = -3$ or $k = 9$

Question 13.

The ellipse $E_1: x^2 + y^2 = 1$ is inscribed in a rectangle R whose sides are parallel to the coordinate axes. Another ellipse E_2 passing through the point $(0, 4)$ circumscribes the rectangle R . The eccentricity of the ellipse is

(a) $\sqrt{2}$

(b) $\sqrt{3}$

(c) 12

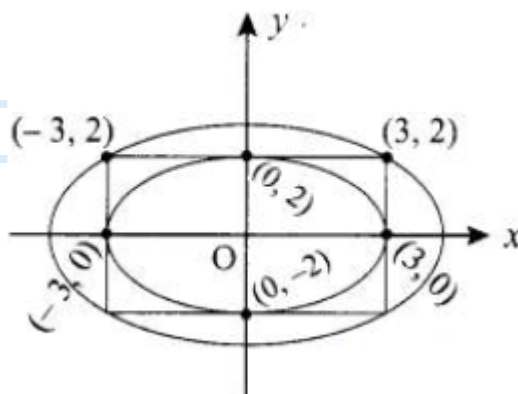
(d) 34

Solution:

(c) 12

Hint:

$e=12$



Question 14.

Tangents are drawn to the hyperbola $x^2 - y^2 = 1$ parallel to the straight line $2x - y = 1$.

One of the points of contact of tangents on the hyperbola is

(a) $(9\sqrt{2}, -1\sqrt{2})$

(b) $(-9\sqrt{2}, 1\sqrt{2})$

(c) $(9\sqrt{2}, 1\sqrt{2})$

(d) $(3\sqrt{3}, -2\sqrt{2})$

Solution:

(c) $(9\sqrt{2}, 1\sqrt{2})$

Hint:

The tangents are parallel to $2x - y = 1$

So equation of tangents will be of the form $2x - y = k \Rightarrow y = 2x - k$

Comparing this equation with $y = mx + c$ we get $m = 2$ and $c = -k$

The given hyperbola is $x^2 - y^2 = 1$

$\Rightarrow a^2 = 9$ and $b^2 = 4$

Now the condition for the line $y = mx + c$ to be a tangent to

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is $c^2 = a^2 m^2 - b^2$ LHS $= c^2 = (-k)^2 = k^2$ RHS $= a^2 m^2 - b^2 = 9(2)^2 - 4 = 32$ Now, LHS = RHS $k^2 = 32 \Rightarrow k = \pm\sqrt{32} = \pm 4\sqrt{2}$

The point of contact is

$$(-a^2mc, -b^2c) = (-a^2m-k, -b^2-k) = (a^2mk, b^2k) = (9(2) \pm 4\sqrt{2}, 4 \pm 4\sqrt{2}) = (92\sqrt{2}, 1\sqrt{2})$$

Question 15.

The equation of the circle passing through the foci of the ellipse $x^2/16 - y^2/9 = 1$ having centre at (0,3) is

(a) $x^2 + y^2 - 6y - 7 = 0$

(b) $x^2 + y^2 - 6y + 7 = 0$

(c) $x^2 + y^2 - 6y - 5 = 0$

(d) $x^2 + y^2 - 6y + 5 = 0$

Solution:

(a) $x^2 + y^2 - 6y - 7 = 0$

The given ellipse is $x^2/16 + y^2/9 = 1$

$$a^2 = 16 \text{ and } b^2 = 9 \Rightarrow e^2 = \frac{a^2 - b^2}{a^2} = \frac{16 - 9}{16} = \frac{7}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$$

$$\text{Now } a = 4 \text{ and } e = \frac{\sqrt{7}}{4} \Rightarrow ae = \sqrt{7}$$

$$\text{Now the foci are } (\pm ae, 0) = (\pm \sqrt{7}, 0)$$

$$\text{Centre} = (0, 3)$$

$$\text{Passing through the foci } = (\pm \sqrt{7}, 0)$$

The distance between centre and foci = radius of the circle

$$\text{ie., } r = \sqrt{(0 \pm \sqrt{7})^2 + (3 - 0)^2} = \sqrt{7 + 9} = \sqrt{16} = 4$$

$$\text{Now Centre} = (0, 3); r = 4$$

$$\text{Equation of the circle is } (x - 0)^2 + (y - 3)^2 = 4^2$$

$$\text{ie., } x^2 + y^2 - 6y + 9 - 16 = 0$$

$$x^2 + y^2 - 6y - 7 = 0$$

Question 16.

Let C be the circle with centre at (1,1) and radius =1. If T is the circle centered at (0,y) passing through the origin and touching the circle C externally, then the radius of T is equal

(a) $\sqrt{3}\sqrt{2}$

(b) $\sqrt{32}$

(c) 12

(d) 14

Solution:

(d) 14

Hint:

Equation of circle is $x^2 + y^2 + 2gx + 2fy + c = 0$

i.e., $(1+y)^2 = (1-y)^2 = 1$

Solving we get $y = 14$

Question 17.

Consider an ellipse whose centre is of the origin and its major axis is along x-axis. If its eccentricity is $\frac{3}{5}$ and the distance between its foci is 6, then the area of the quadrilateral inscribed in the ellipse with diagonals as major and minor axis of the ellipse is

(a) 8

(b) 32

(c) 80

(d) 40

Solution:

(d) 40

Hint:

$$e = \frac{3}{5} \Rightarrow ae = 3 \Rightarrow a^2 e^2 = 9 \Rightarrow a^2 \left(\frac{9}{25}\right) = 9 \Rightarrow a^2 = 25 \Rightarrow a = 5$$

$$\text{Now } b^2 = a^2(1 - e^2) \Rightarrow b^2 = 25\left(1 - \frac{9}{25}\right) = 16 \Rightarrow b = 4$$

$$\text{Now area of the quadrilateral is } \frac{1}{2}(d_1 \times d_2) = \frac{1}{2}(10 \times 8) = 40$$

Question 18.

Area of the greatest rectangle inscribed in the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is

(a) $2ab$

(b) ab

(c) $\sqrt{ab}$

(d) ab

Solution:

(a) $2ab$

Hint:

$$\text{Area of the greatest rectangle is } (\sqrt{2}a)(\sqrt{2}b) = 2ab$$

Question 19.

An ellipse has OB as semi minor axes, F and F' its foci and the angle FBF' is a right angle. Then the eccentricity of the ellipse is

- (a) $1\sqrt{2}$
- (b) 12
- (c) 14
- (d) $1\sqrt{3}$

Solution:

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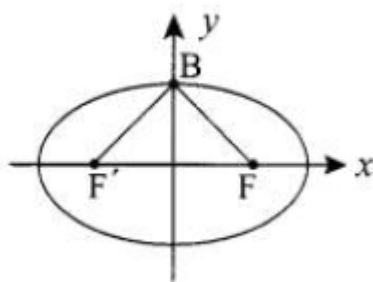
- (a) $1\sqrt{2}$

Hint:

Here $F_1B + FB = FF'$

ie., $(\sqrt{a^2e^2+b^2})^2 + (\sqrt{a^2e^2+b^2})^2 = (2ae)^2$

$\Rightarrow e = \frac{1}{2}$



Question 20.

The eccentricity of the ellipse $(x-3)^2 + (y-4)^2 = y^2/9$ is

- (a) $\sqrt{32}$
- (b) 13
- (c) $13\sqrt{2}$
- (d) $1\sqrt{3}$

Solution:

- (b) 13

Hint:

The given equation is of the form $FP^2 = e^2 PM^2$

ie., $(x-3)^2 + (y-4)^2 = 19(y^2/9) \Rightarrow e^2 = 19 \Rightarrow e = \sqrt{19}$

Question 21.

If the two tangents drawn from a point P to the parabola $y^2=4x$ are at right angles then the locus of P is

- (a) $2x+1=0$
- (b) $x=-1$
- (c) $2x-1=0$
- (d) $x=1$

Solution:

- (b) $x=-1$

Hint:

When the tangents are drawn from a point on the directrix are at right angles.

So equation of directrix to $y^2=4x$ is $x=-1$

Question 22.

The circle passing through $(1,-2)$ and touching the axis of x at $(3,0)$ passing through the point

- (a) $(-5,2)$
- (b) $(2,-5)$
- (c) $(5,-2)$
- (d) $(-2,5)$

Solution:

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- (c) $(5,-2)$

Hint:

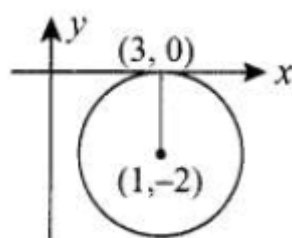
Let the centre be $(3, h)$ $r=k$

Equation of the circle is

$$(x-3)^2+(y-h)^2=r^2=k^2$$

It passes through $(1,-2) \Rightarrow k^2=8$

Substituting $(5,-2)$ the equation satisfies.



Question 23.

The locus of a point whose distance from $(-2,0)$ is 23 times its distance from the line $x=-92$ is

- (a) a parabola
- (b) a hyperbola
- (c) an ellipse
- (d) a circle

Solution:

(c) an ellipse

Hint:

Here $e=23 < 1$

So the conic is an ellipse

Question 24.

The values of m for which the line $y=mx+2\sqrt{5}$ touches the hyperbola $16x^2-9y^2=144$ are the roots of $x^2-(a+b)x-4=0$, then the value of $(a+b)$ is

- (a) 2
- (b) 4
- (c) 0
- (d) -2

Solution:

(c) 0

Hint:

$$16x^2-9y^2=144 \Rightarrow 16x^2-9y^2=144 \text{ ie., } x^2-9y^2=16$$

Here $a^2=9; b^2=16$

The condition for the line $y=mx+c$ to be a tangent to $x^2/a^2-y^2/b^2=1$ is

$$c^2=a^2m^2-b^2$$

Here the given line is $y=mx+2\sqrt{5}$

$$\Rightarrow m=m \text{ and } c=2\sqrt{5}$$

The condition is $c^2=a^2 m^2-b^2$

$$\text{ie., } (2\sqrt{5})^2=9(m^2)-16$$

$$\Rightarrow 9 m^2=20+16=36 \Rightarrow m^2=4$$

$$\text{i.e., } m=\pm 2$$

$$a=2, b=-2 \text{ (say)}$$

$$\text{So } a+b=2-2=0$$

Question 25.

If the coordinates at one end of a diameter of the circle $x^2+y^2-8x-4y+c=0$ are $(11,2)$, the coordinates of the other end are

(a) $(-5,2)$

(b) $(-3,2)$

(c) $(5,-2)$

(d) $(-2,5)$

Solution:

(b) $(-3,2)$

Hint:

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Centre $= (4,2)$ one end of diameter $(11,2)$ $(4,2)$ (x,y)

Let the other end be (x,y)

Now centre = mid point of the end points of diameter.

$$\Rightarrow (4,2) = \left(\frac{11+x}{2}, \frac{2+y}{2} \right)$$

$$(11, 2) \quad (4, 2) \quad (x, y)$$

$$\Rightarrow 11+x=8 \quad 2+y=4 \Rightarrow x=-3 \quad y=2$$

Additional Problems

Choose the most appropriate answer:

Question 1.

The parabola $y^2=4ax$ passes through the point $(2,-6)$, the length of its latus rectum is

(a) 9

(b) 16

(c) 18

(d) 6

Solution:

(c) 18

Hint:

The given parabola is $y^2 = 4ax$

Since it passes through $(2, -6)$

$$\therefore (-6)^2 = 4a(2) \Rightarrow 36 = 8a \Rightarrow a = \frac{36}{8} = \frac{9}{2}$$

Length of the latus rectum $= 4a = 4\left(\frac{9}{2}\right) = 18$

Question 2.

The vertex of the parabola $x^2 + 12x - 9y = 0$ is

(a) $(6, -1)$

(b) $(-6, 4)$

(c) $(6, 4)$

(d) $(-6, -4)$

Solution:

(d) $(-6, -4)$

Hint:

Given parabola is $x^2 + 12x - 9y = 0$

$$\Rightarrow (x^2 + 12x + 36) - 36 - 9y = 0 \Rightarrow (x+6)^2 - 9y - 36 = 0 \Rightarrow (x+6)^2 = 9(y+4) \therefore \text{Vertex} = (-6, -4)$$

Question 3.

The length of the major axis of an ellipse is three times the length of minor axis, its eccentricity is

(a) 13

(b) $\frac{1}{\sqrt{3}}$

(c) $\frac{1}{\sqrt{2}}$

(d) $\frac{2\sqrt{3}}{3}$

Solution:

(d) $\frac{2\sqrt{3}}{3}$

Hint:

Length of major axis $= 2a$ and Length of minor axis $= 2b$

$$2a = 3(2b) \Rightarrow a = 3b \text{ Now } b^2 = a^2(1 - e^2) \Rightarrow e^2 = \frac{a^2 - b^2}{a^2} = \frac{9b^2 - b^2}{9b^2} = \frac{8}{9} \Rightarrow e = \frac{2\sqrt{2}}{3}$$

Question 4.

The eccentricity of the ellipse $9x^2 + 4y^2 = 36$ is

(a) $\frac{\sqrt{5}}{3}$

(b) $\sqrt{35}$

(c) $\sqrt{35}$

(d) $\sqrt{53}$

Solution:

(d) $\sqrt{53}$

Hint:

Given equation of the ellipse $9x^2+4y^2=36$

On dividing both the sides by 36 , we get

$x^2/4+y^2/9=1$. Here $a < b$. $\therefore$ Eccentricity is given by

$$a^2=b^2(1-e^2) \Rightarrow a^2/b^2=1-e^2 \Rightarrow e^2=1-a^2/b^2=b^2-a^2/b^2=9-4=5/9$$

$$e=\sqrt{5/3}$$

Question 5.

S and T are the foci of the ellipse $x^2/a^2+y^2/b^2=1$ and B is an end of the minor axis. If STB is an equilateral triangle, the eccentricity of the ellipse is

(a) $1/\sqrt{2}$

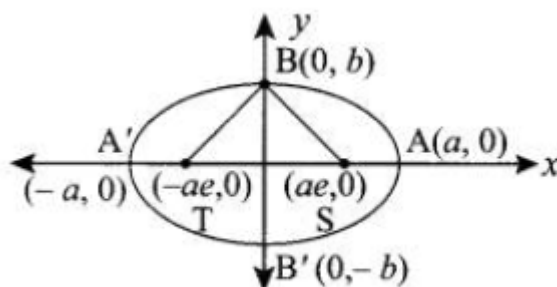
(b) $1/3$

(c) $1/2$

(d) $\sqrt{3}/2$

Solution:

(c) $1/2$



Hint:

S and T are the foci of the ellipse

$x^2/a^2+y^2/b^2=1$. $\therefore S=(ae,0)$ and $T=(-ae,0)$ and $B=(0,b)$. $\therefore$ STB is an equilateral triangle

$$\therefore ST^2=TB^2 \Rightarrow (ae+ae)^2+(0-0)^2=(-ae-0)^2+(0-b)^2 \Rightarrow (2ae)^2=a^2e^2+b^2 \Rightarrow 4a^2e^2=a^2e^2+b^2 \Rightarrow 3a^2e^2=b^2$$

$$[b^2=a^2(1-e^2)] \Rightarrow 4a^2e^2=a^2 \Rightarrow e^2=1/4 \Rightarrow e=1/2$$

Question 6.

The sum of the focal distances from any point of the ellipse $9x^2+16y^2=144$ is

- (a) 32
- (b) 18
- (c) 16
- (d) 8

Solution:

- (d) 8

Hint:

We have, $9x^2+16y^2=144$

$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1$ (Dividing both sides by 144)

Here $a^2=16, b^2=9 \Rightarrow a=4, b=3$

Length of major axis $= 2a = 2(4) = 8$

Since the sum of the focal distances from any point on the ellipse is equal to its major axis

$\therefore$ Required sum $= 8$

Question 7.

If the eccentricities of two ellipse $\frac{x^2}{16} + \frac{y^2}{25} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ are equal then $ab =$

- (a) 513
- (b) 613
- (c) 135
- (d) 136

Solution:

- (c) 135

Hint:

Let e_1 and e_2 be the eccentricities of the ellipses $\frac{x^2}{16} + \frac{y^2}{25} = 1$ and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ respectively

$\therefore$ The eccentricities of the two ellipses are equal,

$\therefore e_1 = e_2 \Rightarrow e_1^2 = e_2^2 \Rightarrow \frac{a^2 - b^2}{a^2} = \frac{a^2 - b^2}{a^2} \Rightarrow \frac{16 - 25}{16} = \frac{a^2 - b^2}{a^2} \Rightarrow 169 - 25169 = a^2 - b^2$

Question 8.

Equation of the hyperbola, whose eccentricity $\frac{3}{2}$ and foci at $(\pm 2, 0)$ is

- (a) $\frac{x^2}{4} - \frac{y^2}{5} = 1$

(b) $x^2 - y^2 = 49$

(c) $x^2 - y^2 = 1$

(d) None of these

Solution:

(a) $x^2 - y^2 = 49$

Hint:

Eccentricity $e = \frac{c}{a} = \frac{3}{2}$

Foci are $(\pm c, 0) = (\pm 3, 0)$ (Given $c = 3$)

$$\therefore \frac{c}{a} = \frac{3}{2} \Rightarrow 2a = 3c \Rightarrow a = \frac{9}{2}$$

Now, $a^2 = \frac{81}{4}$ and $b^2 = 209$

$$\therefore \text{Equation of hyperbola is } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{\frac{81}{4}} - \frac{y^2}{209} = 1 \Rightarrow 4x^2 - 9y^2 = 81 \Rightarrow x^2 - \frac{9}{4}y^2 = \frac{81}{4}$$

Question 9.

If e_1 is the eccentricity of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ and if e_2 is the eccentricity of the hyperbola $9x^2 - 16y^2 = 144$, then $e_1 e_2$ is

(a) $\frac{16}{25}$

(b) 1

(c) Greater than 1

(d) Less than 12

Solution:

(b) 1

Hint:

Equation of ellipse is $\frac{x^2}{25} + \frac{y^2}{9} = 1$

Here, $a^2 = 25, b^2 = 9$

$$\text{Now, } b^2 = a^2(1 - e_1^2) \Rightarrow 9 = 25(1 - e_1^2)$$

$$\Rightarrow e_1^2 = \frac{16}{25}; e_1 = \frac{4}{5}$$

Equation of hyperbola is $\frac{x^2}{16} - \frac{y^2}{9} = 1$

Here,

$$a^2 = 16, b^2 = 9$$

$$b^2 = a^2(e_2^2 - 1) \Rightarrow 9 = 16(e_2^2 - 1)$$

$$\Rightarrow e_2^2 - 1 = \frac{9}{16} \Rightarrow e_2^2 = \frac{25}{16}; e_2 = \frac{5}{4} \therefore e_1 e_2 = \frac{4}{5} \times \frac{5}{4} = 1$$

Question 10.

The point of intersection of the tangents at $t_1 = t$ and $t_2 = 3t$ to the parabola $y^2 = 8x$ is

(a) $(6t^2, 8t)$

(b) $(8t, 6t^2)$

(c) $(t^2, 4t)$

(d) $(4t, t^2)$

Solution:

(a) $(6t^2, 8t)$

Hint:

Point of intersection of the tangents at t_1 and t_2 to $y^2=4ax$ is $[at_1t_2, a(t_1+t_2)]$

Here, $t_1=t, t_2=3t, a = \frac{8}{4} = 2$

So $at_1t_2 = 2(t)(3t) = 6t^2$

$a(t_1+t_2) = 2(t+3t) = 8t \therefore$ Point $= (6t^2, 8t)$

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